Department of Homeland Security (DHS)

United States Coast Guard (USCG)

Homeland Security Cutter – Light Icebreaker (HSC-L)

Request for Proposal

(RFP)

STATEMENT OF WORK

(SOW)

18 MAR 2026

Revision DRAFT

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RECORD OF CONTRACT CHANGES

Version	Date	Purpose
Orig	05/XX/2026	Initial Release

C.1 [Orig] Background

(a) The U.S. Coast Guard maintains and operates icebreaking vessels to promote safe navigation and facilitate commerce in U.S. waters. These missions originate in Executive Order 7521 (21 Dec 1936), which directs the U.S. Coast Guard (USCG) to provide domestic icebreaking in ice-covered waterways to facilitate safe commercial navigation and prevent ice-related flooding. These missions prove critical to the Northeast region, where the economy depends on key shipping channels to maintain the transit of over 100 million tons of vital commodities annually.

The Homeland Security Cutter (HSC) program intends to develop a materiel solution to recapitalize the Coast Guard's light icebreaking capability in Northeast region. The current 65-ft WYTL class of light domestic icebreakers has exceeded 50 years of service. The fleet can no longer meet mission or regulatory demands. The HSC-Light Icebreaker (HSC-L) program plans to acquire seven new vessels to carry on the light icebreaker mission.

C.2 [Orig] General Statement of Work

(a) [Orig] The HSC-L Program intends to acquire seven HSC-L boats collectively called HSC-L under this contract, which are to be fully operational and sustainable over a projected 30-year life cycle. This solicitation seeks a Shipbuilder to produce HSC-Ls that provide long term supportability and affordability while complying with USCG technical and management requirements.

(b) [Orig] The HSC-L contract design is provided. The Contractor shall further develop this design product to be production-ready. The production-ready design shall have the following primary operational capability: icebreaking to promote safety and facilitate commerce in U.S. waters. Secondary capabilities shall include; establish, maintain, and operate aids to navigation; position and perform maintenance on Federal floating aids to navigation on inland waterways; position and perform maintenance on river buoys; position and construct beacons; perform maintenance on fixed aids to navigation; transmit/receive marine safety information; provide lift capability; escort vessels; provide command and control; conduct communications; employ sensors; conduct search & rescue.

(c) [Orig] This Contract encompasses the technical, procurement, and other tasks necessary to develop a production-ready design from the USCG contract design and produce HSC-Ls.

(d) [Orig] The term "lead boat" within the context of this contract refers to the first HSC-L produced under this contract. The lead boat is also referred to as HSC-L-1.

(e) [Orig] Reference document effectivity shall be in accordance with the Section J Attachment J-1, "Technical Specification" Section 042.4.

(f) [Orig] Equipment and Design Standardization. The following systems and equipment on HSC-Ls are standardized in support of operability and sustainability.

1. [Orig] The Scalable Integrated Navigation System - 2 (SINS-2).
2. [Orig] The Automatic Identification System - 2 (AIS-2).
3. [Orig] The supplier of the equipment and sensors with comprise SINS-2 and AIS is:

   Teledyne FLIR Maritime US Inc.
   110 Lowell Road, Hudson, NH 03051, USA
   603.324.7900

(g) [Orig] Supply Chain Risk Management. The Contractor shall implement supply chain assurance to reduce risk of counterfeit and non-conforming supplies and ensure resilience of the supply chain. The Contractor shall source critical components only from original manufacturers or authorized suppliers and maintain documentation of authenticity for such.

(h) [Orig] Delivery and Acceptance. Delivery of each HSC-L by the Contractor, and Acceptance of each HSC-L by the USCG comprise a single contract execution event which will occur at USCG Yard as specified in Section F clause titled "Lead and Follow Boats". The words Delivery and Acceptance in this contract are used interchangeably and refer to the same event.

(i) [Orig] USCG will conduct a Post-Delivery Availability at the USCG Yard, Baltimore, MD. Off-hour time will be made available for Shipbuilder employees or representatives to correct trial cards and provide warranty repairs.

(j) [Orig] The Contractor shall incorporate into their production, the selected equipment identified in the proposal and incorporated into the contract as Section J, Attachment J-14, Major Equipment List. Changes to the Major Equipment List must be requested by the contractor and approved by the Government prior to implementation. Government approval of changes to the Major Equipment List does not alleviate the contractor's responsibility to meet all other requirements of this contract.

C.2.1 [Orig] American Bureau of Shipping (ABS) Communications

(a) [Orig] If required, the Contractor shall authorize direct communication between the Coast Guard and ABS for compliance, progress and reporting issues. Any communication with ABS by either the shipbuilder or the Coast Guard shall be shared by all parties.

C.2.2 [Orig] Contract Award and Communications

(a) [Orig] The Coast Guard will provide an Integrated Data Environment (IDE).

(b) [Orig] All official contract execution correspondence shall be accomplished via the IDE. The Contractor shall obtain an appropriate number of Common Access Cards (CAC) or public key infrastructure (PKI) tokens as per the procedures in Section I clauses titled "Facility and Computer Access" to execute official contract correspondence in the IDE.

(c) [Orig] Contract data deliverables shall be submitted via the IDE unless otherwise annotated in the Data Item.

(d) [Orig] The Effective Date of Contract Award (EDCA) is established in the SF33.

(e) [Orig] If the IDE is inaccessible, deliverables shall be submitted by email on or before their contractual due dates to the USCG Contracting Officer at the address specified in Section F clause titled "Data Deliverables".

C.2.3 [Orig] Design and Engineering (CLIN -001)

(a) [Orig] The Contractor shall develop the production-ready design of HSC-L-1 from the USCG Contract Design through Preliminary Design Review (PDR), Critical Design Review (CDR), and Production Readiness Review (PRR). Design and Engineering also includes:

- Systems interface and integration;
- Requirements testing and verification;
- Production design and preparations through completion of final transitional designs;

- Human systems integration; and
- Mockups.

(b) [Orig] The Contractor shall execute technical reviews in accordance with Attachment J-2, "Design and Readiness Review Requirements and Criteria".

(c) [Orig] The Contractor shall prepare and deliver design data in accordance with Attachment J-4, "Contract Data Requirements List".

(d) [Orig] All data deliveries and milestones during design will align with the Section F clause titled "Period of Performance (POP) / Delivery Schedule" and Attachment J-4, "Contract Data Requirements List". The Date of Contract Award shall be the baseline start date for contract execution unless otherwise specified by the Contracting Officer.

(e) [Orig] Technical data associated with preliminary, functional, and transitional design shall be delivered.

C.2.4 [Orig] Design and Engineering Changes (CLIN -001)

(a) [Orig] The Contractor shall implement Government-initiated and Contractor-proposed change orders to the design under CLIN -001.

(b) [Orig] Change orders can only be implemented after the change order has been authorized by the Contracting Officer via a contract modification.

(c) [Orig] Design and Engineering Changes are applicable only to CLIN -001 Design and Engineering. The pricing impact associated with other CLINs under this Contract shall be incorporated into the Pricing Table at the time of the bi-lateral contract modification, which each Pricing Table being labeled to correspond with the modification number. The Pricing Table shall reflect submitted and approved Design and Engineering Change costs for CLIN -001 in addition to costs of production CLINs affected by changes on both an individual and cumulative scale.

(d) [Orig] Costs other than those allocated under CLIN -001 will be incorporated into Attachment J-XXX of the Contract prior to commencement of production.

(e) [Orig] The steps below outline the general procedures for the Engineering Change Proposal (ECP) approval process for each change order.

1. [Orig] Both the Contractor and Government can propose change orders.

    a. [Orig] For the Government proposed change orders, the Government will provide an Engineering Change Request (ECR) to the Contractor. The ECR is a request for change that is reviewed and approved by the Government led Configuration Control Board to proceed to the ECP process.

    b. [Orig] For Contractor proposed change orders, an ECP can be generated by the Contractor without a preceding ECR.

2. [Orig] The Contractor shall submit an ECP to the Government, which will be reviewed internally by the Government.

3. [Orig] The Contractor and Government will enter negotiations to align on scope and cost of the ECP.

4. [Orig] After an ECP has been accepted by the Government, the Government will issue a bi-

lateral modification to update language within the contract and attachments, as necessary.

5. [Orig] Once modification is issued, the Contractor shall proceed immediately in accordance with the scope of the modification.

6. [Orig] Invoicing shall adhere to requirements set forth in G.16 CONTRACT ADMINISTRATION FOR DESIGN AND ENGINEERING CHANGES.

(f) [Orig] All implemented Design and Engineering Changes shall adhere to the same procedures and scope outlined in CLIN -001 Design and Engineering.

C.2.5 [Orig] Long Lead Time Materials (SLIN -002AA)

(a) [Orig] The Contractor shall procure long lead time materials (LLTM) identified in the Section J Attachment J-7, "Long Lead Time Material (LLTM) and Insurance Spares List." LLTM is Contractor Furnished Equipment. USCG acceptance of LLTM is concurrent with boat delivery. Warranty for all LLTM shall begin at boat delivery.

(b) [Orig] A Delivery Order for LLTM SLINs for HSC-L-1 in Ordering Period 1 shall not be issued until after successful completion of Preliminary Design Review or when authorized by the Contracting Officer.

(c) [Orig] All responsibility for performance, timeliness, and quality of LLTM other than directed standardized equipment remains with the Contractor. The Contractor shall maintain the condition of LLTM through proper storage and care.

(d) [Orig] LLTM for each follow boat is subject to Economic Price Adjustment for materiel costs as itemized per the procedure in Section H clause titled "Economic Price Adjustment".

C.2.6 [Orig] Production of Lead Boat; HSC-L-1 (SLIN -002AB)

(a) [Orig] Placement of these production orders is predicated upon successful completion of the Critical Design Review, a Production Readiness Review, and Contracting Officer direction.

(b) [Orig] The Contractor shall construct the lead HSC-L in accordance with the production design developed by the Contractor.

(c) [Orig] The lead boat shall be built, tested and verified to meet the requirements of the Section J attachment, Section J-1, "Technical Specification". Any requirement not in compliance shall nonetheless be satisfied by the Contractor unless specifically stated in the contract to be the responsibility of the Government.

(d) [Orig] The Contractor shall be responsible for procurement, installation, integration and testing of all Contractor Furnished Equipment (CFE) including software.

(e) [Orig] The lists of Government Furnished Information, which the USCG will furnish, are included in Section J.

(f) [Orig] The Contractor shall be responsible for the procurement and outfit of onboard spares, repair parts, equipage, tools and support equipment/documentation for all Contractor Furnished Equipment.

(g) [Orig] The Contractor shall provide system stock initial logistics spares, including depot level maintenance special tools, for each HSC-L and deliver them as specified in Section F clause titled "System Stock, Consumables, Training Material, and Other Associated Items".

(h) [Orig] The Contractor shall provide spare and consumable parts for urgent/emergent preventive and/or corrective maintenance. These orders may require expedited handling. Spare and consumable parts will be delivered directly to a boat as specified in the order placed in accordance with the Section H clause titled “Orders”.

(i) [Orig] The Contractor shall provide outfitting, comprised of Storeroom Items (SRI), Operating Space Items (OSI), General Use Consumables (GUC), and Allowance Equipage List (AEL) items, and install these items in each boat prior to delivery. The Contractor shall outfit each boat before delivery in accordance with industry standards.

(j) [Orig] The Contractor shall prepare and integrate all plans and schedules required for construction of the lead HSC-L.

(k) [Orig] The Contractor shall be responsible for overall project management, subcontractor management, technical direction, coordination and administrative work to ensure that all contractual requirements are properly accomplished.

(l) [Orig] The Contractor shall conduct required verification, testing and trials (including but not limited to Acceptance Trials).

(m) [Orig] The Contractor shall provide access during production as per the procedures in Section H clauses titled “Access to Contractor’s Facility” and “Access to the Boats” and in accordance with Section C.3.087, and Section C.3.097.

(n) [Orig] The Contractor shall deliver each HSC-L complete in all respects, including documentation and certifications, with discrepancies corrected.

(o) [Orig] The Contractor shall prepare and deliver data in accordance with Attachment, J-4, “Contract Data Requirements List.”

(p) [Orig] Technical data as described in paragraph C.2.8 below for each lead HSC-L shall be delivered (CLIN -003).

(q) [Orig] The Contractor shall provide a Warranty in accordance with the period of performance established in Section F clause titled “Warranty SLINs (XX)” and Section H clause titled “Warranty”.

(r) [Orig] Training Courses for crew members of each HSC-L shall be provided (CLIN -004) as described in paragraph C.2.10 below.

(s) [Orig] The Contractor shall provide 200 hours of subject matter expert (SME) time to educate and inform Government and Government-contracted personnel related to then analysis, design, development and implementation of Familiarization Training (to include underway training) and review related materials. SME access will be utilized by Government and contracted training personnel to conduct training analysis on shipboard systems. Further shipbuilder SMEs may be asked to verify the design and development of training materials for validation of content.

C.2.7 [Orig] Follow Boat Production (SLIN -002AB)

(a) [Orig] The Coast Guard plans to order for production each follow-boat listed in Section B, and to authorize production of at least the number of HSC-L listed in each ordering period. Design Reviews at noted in paragraph C.3.068 may be consolidated or waived based upon agreement between the Contractor, Contracting Officer and/or COR.

(b) [Orig] Each follow HSC-L is subject to Economic Price Adjustment for material costs as itemized per the procedure in Section H clause titled "Economic Price Adjustment".

(c) [Orig] Each requirement stated for the HSC-L lead boat in paragraphs C.2.6(c) through C.2.6(s) shall also apply to each follow-HSC-L.

(d) [Orig] Technical data as described in paragraph C.2.8 below for each follow HSC-L shall be delivered (CLIN -003 SLINS).

(e) [Orig] The Contractor shall provide a Warranty in accordance with the period of performance established in Section F clause titled "Warranty SLINs (XX)" and Section H clause titled "Warranty".

(f) [Orig] Training Courses for crew members of each follow HSC-L shall be provided as described in paragraph C.2.10 below (CLIN -004 SLINs).

C.2.8 [Orig] Technical Data Submissions (CLIN -003)

(a) [Orig] The Contractor shall deliver Data Rights as listed in Section J, Attachment J-9A, "Technical Data Rights Tables".

C.2.9 [Orig] Optional Technical Data Rights (SLINs -003AA, -003AB, -003AC)

(a) [Orig] The Contractor shall deliver Optional Technical Data Rights as listed in Section J, Attachment J-9B, "Restricted Data Technical Data and Software Matrices" as ordered by the Government.

(b) [Orig] Orders will be placed in accordance with the clause in Section H clause titled "Orders (Fixed Price)" and at the prices provided by the Contractor in Section J, Attachment J-9B, "Restricted Data Technical Data and Computer Software Matrices."

C.2.10 [Orig] Training (CLIN -004)

(a) [Orig] The Contractor shall provide crew familiarization (FAM) and original equipment manufacturer training to USCG personnel for each HSC-L delivered.

C.2.10.1 [Orig] Crew Familiarization Training (SLIN -004AA)

(b) [Orig] The Contractor shall provide five days of crew familiarization (FAM) within two days of the boat being delivered unless otherwise arranged between the Contractor and USCG.

1. [Orig] The FAM shall be designed so that the crew will be able to operate all controls, navigation, and communication equipment to safely pilot the boat.

2. [Orig] The FAM shall concentrate on the safe use of the boat, basic operation of all boat controls and equipment, and maintenance.

3. [Orig] The FAM shall be provided at the acceptance location.

4. [Orig] The prescribed time for FAM shall cover both operator and maintainer FAM requirements. Up to 10 operators and up to 10 maintainers shall receive FAM during each session. Operator and maintainer FAM sessions may be run concurrently for time efficiency. The FAM may include classroom sessions in addition to actual operation of the boat.

5. [Orig] FAM shall be performance-based and shall be conducted using operational equipment installed onboard the delivered boat.

6. [Orig] FAM training instructors shall provide step-by-step demonstrations of system/equipment operation and maintenance that shall be immediately followed by a time during which students reenact and perform the identical sequence of tasks.

7. [Orig] Operator and Maintainer FAM shall address all topics included in the BIB and OEM manuals (as applicable). The BIB and OEM manuals shall be used as a FAM aid.

(c) [Orig] At a minimum, operator FAM shall include, but not be limited to:

1. [Orig] Basic operation of all boat systems, sub-systems, equipment, operational characteristics, and parameters such as basic construction features, draft, turn radius, stopping distance, maximum sea states, and maximum safe speeds.

2. [Orig] Controls for boat handling, deck equipment, engines, electronics, fuel, bilge, fendering, and propulsion.

3. [Orig] Start-up and shut-down procedures.

4. [Orig] Warnings.

5. [Orig] Casualty control and emergency procedure for each action in the BIB and OEM manuals.

6. [Orig] Safe piloting, to include navigating at speed, holding station, and slow speed maneuvering alongside a pier or ship.

7. [Orig] Eight hours underway with the USCG operating the boat under Contractor supervision.

(d) [Orig] At a minimum, maintainer FAM shall cover, but not be limited to, organizational preventative and corrective maintenance for the following systems: propulsion, engines, electrical, electronics, steering, and hotel systems. The Contractor shall demonstrate the use of all special tools provided.

(e) [Orig] The Contractor shall provide a syllabus that includes, at a minimum, an agenda listing the topics and approximate time devoted to each **[CDRL]**.

(f) [Orig] A copy of all materials used for the FAM shall be provided to the USCG upon completion of FAM.

C.2.10.2 [Orig] Original Equipment Manufacturer (OEM) Training (SLIN -004AB)

(g) [Orig] The Contractor shall procure and coordinate delivery of available OEM training courses related to the equipment on each HSC-L for the HSC-L crews for each HSC-L delivered. Training shall cover Organizational level (O-level) equipment maintenance, troubleshooting and operation.

1. Minimum equipment/system training requirements are provided below.

- Propulsion Engine.
- Reduction Gear.
- Generator Engine(s).
- Electrical Distribution Panels.
- Steering System.
- Integrated Bridge System.
- Machinery Plant Control and Management System.
    - Propulsion Control
    - Integrated Bridge System

- Deck Equipment and Crane.

2. [Orig] The Government will select the required courses from this list. The Contracting Officer will direct the Contractor to provide the courses required per each HSC-L delivered.

3. [Orig] This will include services and courses from Original Equipment Manufacturer authorized trainers for major equipment in accordance with **CDRL**. Training shall be conducted at the Contractor's or Subcontractor's facility as appropriate. The Contractor shall obtain prior USCG approval before arranging any COTS training in a foreign country.

C.2.11 [Orig] Deep Insurance Spares (CLIN -005)

(a) [Orig] The Contractor shall provide Deep Insurance Spares to replace major equipment that becomes damaged beyond repair and shall be comprised of selected Long Lead Time Material components. The government anticipated insurance spare list is provided in Attachment J-7, "Long Lead Time Material (LLTM) and Insurance Spares List."

(b) [Orig] The Government may order Deep Insurance Spares in Attachment J-7, "Long Lead Time Material (LLTM) and Deep Insurance Spares List" at the unit prices. Deep Insurance Spares shall be delivered as required by Section F clause titled "Insurance Spares".

C.2.12 [Orig] Operational Test Support (CLIN -006)

(a) [Orig] Operational Test Support as described in paragraph C.3.080 and C.3.092 below shall be provided for each boat (CLIN -006).

C.2.13 [Orig] Boat Transport to Delivery (CLIN -007)

(a) [Orig] Each HSC-L ordered shall be prepared for sea and delivered by a qualified master and crew to USCG Yard, Baltimore, MD. A limited number of USCG crew will accompany the HSC-L during the delivery.

C.2.14 [Orig] Special Studies (CLIN -008)

(a) [Orig] If ordered, the Contractor shall conduct special studies and analyses in support of this contract for non-warranty and/or Government-required changes.

(b) [Orig] The Contractor shall provide engineering services in support of the boat(s) to accomplish, in a timely manner, activities including but not limited to design activities in support of fielding the HSC-Ls as well as corrections for minor repairs and deficiencies that are deemed to be the Government's responsibility. Requirements under this CLIN might include, but are not limited to, the correction of problems with Government Furnished Equipment (GFE), Government requirements identified by Acceptance Trial cards, design of homeport specific brows, and the development of various engineering products/drawings. The requirements associated with this CLIN are over and above those specified in other contract CLINs.

(c) [Orig] Orders under this CLIN will be placed by a Coast Guard approved Technical Instruction (TI) in accordance with the Section H clause titled "Orders (Fixed Price)". Prior to the placement of an order, the contractor shall submit, as requested, a draft TI for approval by the Coast Guard detailing the specific effort to be accomplished, the period of performance covered by the TI, the FFP cost proposal for the work and the list of deliverables that will be submitted.

(d) [Orig] Travel billed in association with these activities will be billed in accordance with Joint Federal Travel Regulations and be approved at least two weeks in advance by the COR.

C.2.15 [Orig] Capital Expenditures (CAPEX) (CLIN -009)

(a) [Orig] The Contractor shall implement facility improvements or process improvement projects to provide cost savings, quality improvements, or schedule efficiencies to the Government for HSC-L construction or other Coast Guard programs.

(b) [Orig] Orders under this CLIN will be placed by a Coast Guard approved Technical Instruction (TI) in accordance with the Section H clause titled "Orders (Fixed Price)" and Section H clause titled "Capital Expenditures (CAPEX)."

C.2.16 [Orig] Display Model (CLIN -010)

(a) [Orig] Display Models. The Contractor shall produce a 1:20 scale display model. The hull, deck arrangement, pilothouse, and all components of the models shall conform accurately to the scaled dimensions of the HSC-L. The model shall be constructed and finished using durable materials that are resistant to humid conditions.

1. [Orig] The model shall be of sufficient detail to accurately reflect the HSC-L including:
    - [Orig] Realistic colors and finishes.
    - [Orig] A hull which is outfitted to accurately reflect the HSC-L including: fendering; propeller; shafting; skegs; and other components of the HSC-L.
    - [Orig] Deckhouse, exterior equipment, and furnishings including: operator's console, seating, deck lockers, cranes, masts, hatches, vents, mooring fittings, tow bitt(s), handrails, non-skid, antennas, lights, outfit including life rings, distress marker lights, lift fittings and boat hooks, and anchor if stored in a visible location.
    - [Orig] Transparent pilothouse windows, allowing visibility to a fully outfitted pilothouse.
    - [Orig] The model shall include USCG markings, including the "U.S. Coast Guard", and hull number. The hull number shall be identical to HSC-L-1.

2. [Orig] The model shall be set on an oak base with a "honey" finish.

3. [Orig] A base shall be provided with port and starboard brass plaques that identify the models and the "US Coast Guard Homeland Security Cutter-Light Icebreaker (HSC-L)" and give the basic dimensions of the HSC-L and the manufacturer's name.

4. [Orig] The model shall be enclosed in a safety plate glass case on a display table with wheels.

5. [Orig] The model shall be provided with an oak display table that is 30 inches (0.76m) high with a width and depth appropriate to contain the model.

6. [Orig] The model, case, and table shall be provided with a crate suitable for shipping. Delivery location is specified in Section F clause titled "Display Models".

C.2.17 [Orig] Ceremonies and Events (CLIN -011)

(a) [Orig] The Contractor shall coordinate and host a keel laying ceremony to mark the formal recognition of the start of HSC-L construction. The Contractor shall provide all necessary logistical support, including but not limited to shipyard venue preparation, event staffing, security, audio/visual equipment, seating, signage, and refreshments. The Contractor shall also support the participation of 200 designated government representatives and guests.

(b) [Orig] The Contractor shall coordinate and host a christening ceremony upon completion of boat construction and prior to delivery. The Contractor shall provide all necessary logistical support, including but not limited to venue preparation, event staffing, security, audio/visual equipment, seating, signage, and refreshments. The Contractor shall also support the participation of 200 designated government representatives and guests.

C.3 [Orig] Specific Statements of Work

(a) [Orig] The following subparagraph numbering corresponds to the Ship Work Breakdown Structure established by the National Shipbuilding Research Program.

C.3.040 [Orig] Program Management

C.3.040.1 [Orig] General

C.3.040.1.1 [Orig] This Statement of Work (SOW) and the associated Specification establish requirements for the design, fabrication, construction, testing, delivery, logistical support, and technical documentation for the HSC-L.

C.3.040.1.2 [Orig] The Contractor shall manage the entire work effort to develop the contract design into a production ready design, fabricate, construct, test, deliver, and support the HSC-L. This shall include ensuring that each HSC-L, logistics, and technical documentation meet the requirements of the SOW and the Specification. The requirements of the SOW and Specification apply to each HSC-L, unless otherwise indicated.

C.3.040.1.3 [Orig] In case of inconsistency between the requirements within documents cited or referenced within the contract, the following order of precedence applies. Note: silence of one document with respect to details shown in another shall not be considered an inconsistency. Additional order of precedence requirements are in Section J Attachment J-1, “Technical Specification”, Section 042.5.

C.3.040.1.3.1 [Orig] The contract requirements.

C.3.040.1.3.2 [Orig] This Statement of Work.

C.3.040.1.3.3 [Orig] The HSC-L System Specification.

C.3.040.2 [Orig] Meetings

C.3.040.2.1 [Orig] General: The Contractor shall host meetings or be present at USCG meetings as required throughout the contract. For all meetings, the Contractor shall provide agendas and minutes unless stated otherwise in the SOW. The Contractor shall prepare and finalize the minutes for all meetings with input from the Contracting Officer’s Representative (COR). Minutes shall include action items, action items’ assignment (i.e., Contractor, USCG), and status. **[CDRL]**

C.3.040.2.2 [Orig] Meetings: The Contractor, Contracting Officer, or COR may request in-person and/or remote meetings to be hosted at the Contractor’s facility or at an alternate location as agreed to by both parties and shall have a video conference for additional off-site USCG representation unless otherwise agreed to by both parties. Meetings shall be requested by providing the Contractor or Contracting Officer/COR notice 30 calendar days in advance or as agreed to by the Contracting Officer.

C.3.040.2.3 [Orig] Identified Formal Meetings, Conferences, and Reviews:

C.3.040.2.3.1 [Orig] Post Award Conference

C.3.040.2.3.1.1 [Orig] The Post Award Conference serves as a kick-off meeting for the contract.

C.3.040.2.3.1.2 [Orig] The Post Award Conference shall be led by the Contractor and include a review of deliverables, warranty terms and conditions, human systems integration, logistics and supply support, the Design and Development Plan, production planning, and any questions, concerns, or clarifications the Contractor requests with the Design Artifacts, the Specification or Statement of Work.

C.3.040.2.3.1.3 [Orig] The conference shall be no more than 5 days in duration, hosted by the Contractor, include sufficient space for 45 Government personnel, and be scheduled to occur within 30 days after contract award at a location agreed upon by both parties.

C.3.040.2.3.2 [Orig] Program Management Reviews (PMR)

C.3.040.2.3.2.1 [Orig] The Contractor shall conduct PMRs at its facility, quarterly, beginning no later than three months after award. The Contractor shall attend each PMR with the appropriate personnel to address issues raised from the agenda items.

C.3.040.2.3.2.2 [Orig] Each PMR shall address the planned efforts for boat design, boat construction, testing, logistics support, human systems integration, management processes, life cycle support, risk management, Long Lead Time Material (LLTM), subcontractor management, procurement schedules, critical material schedules, and shall include identification and resolution of technical and management issues affecting schedules, cost, and performance and program risk. Each PMR shall also provide an update regarding the exact count of active employees by trade within the shipyard, planned hiring activities to maintain staffing, and other risks, opportunities, and concerns. **[CDRL]**

C.3.040.2.3.2.3 [Orig] Each PMR shall include splinter sessions to address issues on the agenda. The Contractor shall coordinate schedules with the Government to run sessions using multiple rooms at the same facility in parallel to address every applicable splinter session topic in a timely fashion.

C.3.040.2.3.3 [Orig] Design Reviews

C.3.040.2.3.4 [Orig] Mockup Reviews

C.3.040.2.3.5 [Orig] Production Readiness Reviews

C.3.040.2.3.6 [Orig] Test Readiness Reviews

C.3.040.2.3.7 [Orig] Post-OT&E Design Reviews

C.3.040.2.3.8 [Orig] Integrated Baseline Reviews

C.3.040.3 [Orig] Data Exchange

C.3.040.3.1 [Orig] All data, drawings, and/or information submitted by the Contractor to the USCG shall be in accordance with section C.2.2, unless otherwise stated in the SOW, Specification, or CDRL to require hard copies to the USCG.

C.3.040.4 [Orig] Integrated Master Schedule (IMS) **[CDRL]**

C.3.040.4.1 [Orig] The Contractor shall develop and maintain an event-driven, Integrated Master Schedule (IMS). The IMS shall show critical development milestone events and accomplishments for the prime and major subcontractors through the end of the contract performance period.

C.3.040.4.2 [Orig] The IMS shall also show all of the following events and milestones:

C.3.040.4.2.1 [Orig] Boat Deliveries

C.3.040.4.2.2 [Orig] Deliverables (e.g., CDRL, data) submission schedule

C.3.040.4.2.3 [Orig] Meetings (e.g., design reviews) and program reviews

C.3.040.4.2.4 [Orig] Schedule of required dates for materials

C.3.040.4.2.5 [Orig] Major milestones

C.3.040.4.2.6 [Orig] Configuration Baselines

C.3.040.4.2.7 [Orig] Contractor's program phases

C.3.040.4.2.8 [Orig] Long-Lead Time Materials (LLTM) order date, as required

C.3.040.4.2.9 [Orig] The IMS shall reflect the baseline start/finish dates as well as the actual start/finish dates of all phases and intervals.

C.3.040.5 [Orig] Integrated Baseline Reviews (IBR): The Contractor shall host an IBR periodically to review the current delivery order and deliverables. IBRs shall be hosted by the Contractor at the Contractor's facility.

C.3.040.5.1 [Orig] IBRs shall be hosted in response to events, or at least annually, and may be held in conjunction with other meetings. Events that require IBRs include: a request by the Contractor, orders that include a new production of HSC-Ls, or an order or contract change that significantly alters the rate or schedule of production.

C.3.041 [Orig] Configuration Management

C.3.041.1 [Orig] General

C.3.041.1.1 [Orig] The Contractor shall be responsible for developing, documenting, and managing configuration of all boats.

C.3.041.1.2 [Orig] The Contractor shall establish unique identifiers for each developed and furnished document, physical hardware/equipment, and software. The unique identifier shall be used consistently between CDRLs.

C.3.041.2 [Orig] Configuration Status Accounting (CSA)

C.3.041.2.1 [Orig] The Contractor shall provide Configuration Status Accounting (CSA) as a means of creating a process to maintain and organize the knowledge base necessary to perform Configuration Management (CM).

C.3.041.2.2 [Orig] The Contractor is responsible for CSA through the establishment of the Product Configuration Baseline (PCB). After the establishment of the PCB, the responsibility for CSA and management of the PCB will be transferred to the USCG.

C.3.041.2.3 [Orig] CSA activities shall meet Section 7 of MIL-HDBK-61B and the CSA program shall address the tasks identified in Figure 8 of MIL-HDBK- 61B. The CSA system shall be accessible, viewable, and maintainable by the USCG through a standard USCG workstation.

C.3.041.3 [Orig] Configuration Baseline Definitions

C.3.041.3.1 [Orig] General: A configuration baseline is a fixed reference configuration established by defining and recording the approved configuration documentation for a system or CI at a milestone event or specified time.

C.3.041.3.2 [Orig] Functional Configuration Baseline (FCB)

C.3.041.3.2.1 [Orig] The Functional Configuration Baseline (FCB) consists of the boat's technical, mission, operational and functional requirements, interoperability and interface characteristics, and design constraints as contained in the Contract Design Artifacts and Specification at contract award, including those items or capabilities that the Contractor offered and were incorporated into the contract.

C.3.041.3.2.2 [Orig] USCG will provide the Contractor with the FCB in the form of the Specification and HSC-L Contract Design artifacts.

C.3.041.3.3 [Orig] The Allocated Configuration Baseline (ACB)

C.3.041.3.3.1 [Orig] The Allocated Configuration Baseline (ACB) is the contractor's design at PMR #1 that formally breaks down the functional requirements identified in the FCB into more detailed functional requirements and interface characteristics.

C.3.041.3.4 [Orig] The Developmental Configuration Baseline (DCB)

C.3.041.3.4.1 [Orig] The Developmental Configuration Baseline (DCB) represents the USCG-accepted "build to" configuration of the first boat. The DCB is the Contractor's responsibility and may be established in phases as a result of a series of detailed design reviews (DDRs) leading up to the Critical Design Review (CDR). When the USCG accepts a deliverable, its contents become part of the DCB. The DCB is established when deliverables associated with the DDRs up to the CDR have been accepted.

C.3.041.3.5 [Orig] Preliminary Product Configuration Baseline (p-PCB)

C.3.041.3.5.1 [Orig] The preliminary Product Configuration Baseline (p-PCB) is established by the Contractor upon USCG acceptance of the technical data that represents the configuration of the first boat. The technical data shall include Contractor-formatted preliminary 3D model (in both native and STEP format), and Contractor-formatted 2D drawing set (PDFs, native, and AutoCAD format) that reflects the as-built configuration.

C.3.041.3.6 [Orig] Product Configuration Baseline (PCB)

C.3.041.3.6.1 [Orig] The Product Configuration Baseline (PCB) is established upon adjudication of the Physical Configuration Audit (PCA). The PCB consists of the complete technical data package as required in SOW Section C.3.085. The USCG will notify the Contractor once the PCB has been established.

C.3.041.3.6.2 [Orig] Once the PCB is established, configuration is under USCG control. The Contractor shall not change or modify the configuration on any boat in production or any boat already in service without approval from the USCG. This includes change or modification during the completion of warranty work.

C.3.041.4 [Orig] Physical Configuration Audit (PCA)

C.3.041.4.1 [Orig] The USCG will conduct a Physical Configuration Audit (PCA) on a production-representative boat to verify conformance with the product drawings.

C.3.041.4.2 [Orig] The boat for the PCA will be designated by the USCG and will incorporate all approved changes identified during operational test and evaluation (OT&E).

C.3.041.4.3 [Orig] SOW Section 085's product drawings will be used for the PCA. The product drawings shall reflect any changes incorporated into production as a result of OT&E. The Contractor shall provide one hard copy set at PCA.

C.3.041.4.4 [Orig] The USCG will conduct an audit at the Contractor's facility, or an alternate facility agreed to by both parties, for the designated HSC-L. The physical boat will be compared to each product drawing and each product drawing will be compared to the physical boat to validate that documented details match. Any discrepancies between the physical boat and the product drawings will be documented in a USCG-generated PCA report.

C.3.041.4.5 [Orig] The PCA is scheduled to take no more than two weeks to complete. A Contractor representative is invited but not required to attend. The boat shall be prepared by the Contractor and accessible to USCG. The Contractor shall provide services to the boat to allow powering up of systems.

C.3.041.4.6 [Orig] The Contractor shall review the report and shall propose a resolution for each discrepancy. Resolution may be in the form of:

C.3.041.4.6.1 [Orig] The drawings are correct and the boats, including those already accepted, those in production, and the PCA boat, shall be corrected.

C.3.041.4.6.2 [Orig] The boat is correct, and the product drawings shall be revised.

C.3.041.4.7 [Orig] The Contractor shall prepare a PCA Resolution Report that shall identify a corrective action for each discrepancy to include anticipated dates when the actions will be completed.

C.3.041.5 [Orig] Configuration Control

C.3.041.5.1 [Orig] Engineering Change Proposal (ECP) - The Contractor can propose changes to the contract requirements, address DMSMS/obsolete, or to the design by submitting a DD Form 1692 supplied as Government Furnished Information (GFI). Engineering Change Proposals (ECPs) shall be in accordance with C.2.4(e).

C.3.041.5.2 [Orig] Request for Variance (RFV) - The Contractor may propose variance from a particular requirement of an item's configuration baseline documentation for a specific hull or number of hulls, or for a specific period, by submitting a Request for Variance (RFV). An RFV differs from an engineering change because it does not require a change to the configuration baseline document but merely documents the authority for the specified hulls and their configuration documentation to differ from the baseline document. A DD Form 1694 supplied as Government Furnished Information (GFI) may be used. The USCG will review RFVs. Upon acceptance in the form of a contract change, the Contracting Officer will authorize the Contractor to implement the RFV.

C.3.041.6 [Orig] Diminishing Manufacturing Sources/Material Shortages (DMSMS) and Obsolescence

C.3.041.6.1 [Orig] The Contractor shall be responsible for addressing Diminishing Manufacturing Sources/Material Shortages (DMSMS) and obsolescence throughout the life of the contract. If DMSMS/obsolescence concerns appear to affect the boat, the Contractor shall inform the USCG as soon as the Contractor becomes aware of the issue.

C.3.041.6.2 [Orig] The Contractor shall be responsible for providing recommended solutions to the USCG, which may include alternate vendors, new sources of supply, or substitute parts.

C.3.041.6.3 [Orig] The Contractor shall bear all costs associated with revising technical documentation and substituting any Configuration Item (CI) when the reason for substitution is that the component is DMSMS/obsolete. This does not apply to USCG specified parts.

C.3.041.6.4 [Orig] Software Development

C.3.041.6.4.1 [Orig] The Contractor shall provide the following for custom software developed for the HSC-L program.

C.3.041.6.4.1.1 [Orig] Interface Design Description (IDD). **[CDRL]**

C.3.041.6.4.1.2 [Orig] System/Subsystem Design Description (SSDD). **[CDRL]**

C.3.041.6.4.1.3 [Orig] Firmware Support Manual (FSM). **[CDRL]**

C.3.041.6.4.1.4 [Orig] Vulnerability and Resolution Report (VRR). **[CDRL]**

C.3.045 [Orig] Care of Boat During Construction

C.3.045.1 [Orig] General

C.3.045.1.1 [Orig] The boat shall be maintained in a clean condition to prevent deterioration or contamination of equipment, machinery, and materials. This shall include routine cleaning and prevention of buildup of condensation that may subject the boat to damage.

C.3.045.1.2 [Orig] Equipment, prefabricated parts, Government-furnished equipment (GFE), and all other items stored by the Contractor for the boat shall be kept clean and protected from the environment and vermin.

C.3.045.1.3 [Orig] Preservatives or protective covers applied by OEM shall be left intact (or replaced if deteriorated, damaged, or removed) until installation of the machinery or equipment on the boat. If removal of the preservative or protective cover is necessary for testing the machinery or equipment prior to installation, the Contractor shall preserve and protect the machinery or equipment, in accordance with the OEM's instructions, until installed.

C.3.045.1.4 [Orig] Preservatives or protective covers on working parts shall be thoroughly removed prior to operation of machinery or equipment.

C.3.045.2 [Orig] Physical Protection

C.3.045.2.1 [Orig] The Contractor shall provide physical protection and access control for the boat and its associated components until acceptance by the USCG.

C.3.045.3 [Orig] Protection of Machinery, Equipment, and Material

C.3.045.3.1 [Orig] The Contractor shall be responsible for the care of all machinery and equipment, whether furnished by the Contractor or by the USCG. A thorough cleaning shall be performed on any equipment, prior to energizing, which may have been subjected to foreign-matter intrusion.

C.3.045.3.2 [Orig] The Contractor shall ensure that all machinery, equipment, and materials are protected during hull blasting evolutions to prevent contamination or unintended damage.

C.3.045.4 [Orig] Protection of the HSC-L During Construction and Trials

C.3.045.4.1 [Orig] All parts of the boat shall be maintained in an undamaged condition during the entire period the boat is in the Contractor's possession. The Contractor shall act as necessary to prevent abnormal wear and damage to the boat and to prevent corrosion or other environmental deterioration during construction, testing, trials, and delivery.

C.3.045.4.2 [Orig] While the boat is in the Contractor's possession, the Contractor is responsible for safeguarding the boat and equipment from damage due to wind, fire, freezing, high temperature, water, and UV exposure.

C.3.045.4.3 [Orig] Temporary covers shall be provided and used during construction to:

C.3.045.4.3.1 [Orig] Protect flooring, machinery, and other equipment requiring protection that is installed aboard the boat.

C.3.045.4.3.2 [Orig] Protect equipment requiring protection from the weather as recommended by the manufacturer.

C.3.045.4.3.3 [Orig] Cover over openings in the boat to protect the interior against damage due to weather.

C.3.045.4.4 [Orig] The Contractor shall use fireproof or fire-resistant covers to prevent damage or possible ignition of equipment or materials due to falling sparks or other potential sources of fire.

C.3.045.4.5 [Orig] During the construction process the Contractor shall protect the boat from damage due to components containing mercury (e.g., fluorescent lights, thermostats, electrical test equipment, etc.).

C.3.045.4.6 [Orig] Inspect and clean the hull and propellers of marine growth and renew all cathodic protection no more than 30 days prior to delivery or acceptance by the USCG.

C.3.045.4.7 [Orig] If the boat is grounded or involved in a collision , the Contractor shall:

C.3.045.4.7.1 [Orig] Immediately notify the Contracting Officer of the incident;

C.3.045.4.7.2 [Orig] Haul the HSC-L from the water and block on shore for inspection, unless otherwise agreed to by the Contracting Officer; and

C.3.045.4.7.3 [Orig] After an inspection, submit an incident report, detailing the incident and any damage found (including pictures).

C.3.045.5 [Orig] Launching, Docking, and Undocking

C.3.045.5.1 [Orig] The Contractor shall properly launch the boat. The Contractor shall be responsible for the safe launching of the boat and its condition of stability when waterborne.

C.3.045.5.2 [Orig] Should there be any evidence that the boat has been damaged during launching, the Contractor shall immediately have the boat hauled from the water and blocked on shore for inspection, unless otherwise agreed to by the Contracting Officer and/or COR.

C.3.045.6 [Orig] The Contractor shall provide continuous cathodic protection while the boat is waterborne.

C.3.068 [Orig] Integration and Engineering

C.3.068.1 [Orig] General

C.3.068.1.1 [Orig] The Contractor shall provide Integration and Engineering to design and produce boats that meet all requirements identified in the Contract.

C.3.068.1.2 [Orig] Integration and Engineering consists of four phases:

C.3.068.1.2.1 [Orig] Phase 1: Design – begins at contract award and concludes at completion of the Critical Design Review (CDR).

C.3.068.1.2.2 [Orig] Phase 2: Development – begins at completion of CDR and concludes at USCG acceptance of the first boat.

C.3.068.1.2.3 [Orig] Phase 3: Operational Test & Evaluation (OT&E) – conducted by the USCG and begins at USCG acceptance of the first boat and concludes after completion of OT&E and initiation of Phase 4: Production

C.3.068.1.2.4 [Orig] Phase 4: Production – begins after USCG completion of OT&E and includes production of all follow-on boats incorporating any changes resulting from OT&E as reviewed and agreed upon in the Post-OT Design Review (Post-OTDR).

C.3.068.1.3 [Orig] The Contractor shall identify when Long-Lead Time Material (LLTM) will need to be ordered by the Contractor to support construction of the first boat in accordance with the Contractor's Integrated Master Schedule.

C.3.068.2 [Orig] Design and Development Plan

C.3.068.2.1 [Orig] The Contractor shall develop a Design and Development Plan (DDP) identifying all design, development, construction, test efforts, and reviews in Phase 1 and Phase 2 of Integration and Engineering.

C.3.068.2.2 [Orig] The DDP shall identify and sequence a series of Design Reviews, including mockup reviews, that represent an incremental approach to design and production of the first boat.

C.3.068.2.3 [Orig] Each DDP activity or Design Review shall identify the relevant Specification sections to be covered during the planned activity or review.

C.3.068.2.4 [Orig] The DDP shall include entrance and exit criteria for each Design Review and describe the technical data content clearly showing the planned delivery schedule of the relevant technical data, manuals, and deliverables.

C.3.068.2.5 [Orig] The DDP shall in a logical progression identify, update and address these design work elements:

C.3.068.2.5.1 [Orig] General arrangement

C.3.068.2.5.2 [Orig] Hull structure and scantling

C.3.068.2.5.3 [Orig] Propulsion system, including resistance, powering, and maneuvering (i.e. station keeping for ATON, crash stop)

C.3.068.2.5.4 [Orig] Control surfaces, rudder sizing, and turning performance

C.3.068.2.5.5 [Orig] Updates to intact stability, damage stability, and towing stability

C.3.068.2.5.6 [Orig] Details of the production design for the electrical power generation, storage, and distribution system

C.3.068.2.5.7 [Orig] Interior lighting system

C.3.068.2.5.8 [Orig] Exterior lighting system

C.3.068.2.5.9 [Orig] Updates and details of the Pilothouse arrangement and seating

C.3.068.2.5.10 [Orig] Updates to the machinery space(s) arrangement

C.3.068.2.5.11 [Orig] Details of the heating, ventilation, and air conditioning systems

C.3.068.2.5.12 [Orig] Mast and antenna configuration

C.3.068.2.5.13 [Orig] Fire detection and suppression system

C.3.068.2.5.14 [Orig] Installation details of the electronics, communications, and navigation systems

C.3.068.2.5.15 [Orig] Propulsion controls, steering system, bow thruster, and autopilot

C.3.068.2.5.15.1 [Orig] Description and human-machine interfaces

C.3.068.2.5.15.2 [Orig] Failure modes and responses

C.3.068.2.5.15.2.1 [Orig] Risk identification and analysis

C.3.068.2.5.15.2.2 [Orig] Failure modes and effects analysis

C.3.068.2.5.16 [Orig] Aft deck arrangement including ATON and towing tackle equipment, configuration and controls

C.3.068.2.5.17 [Orig] Galley, berths, and habitability details.

C.3.068.2.5.18 [Orig] Human factors engineering for operators and maintainers

C.3.068.2.5.19 [Orig] System reliability

C.3.068.2.5.20 [Orig] Noise and vibration

C.3.068.2.5.21 [Orig] Coating system, including a preservation plan in accordance with SFLC Std Spec 6310.

C.3.068.3 [Orig] Design Reviews

C.3.068.3.1 [Orig] General

C.3.068.3.1.1 [Orig] A series of Design Reviews shall be held by the Contractor to demonstrate the technical documentation for contract compliance and requirements traceability. There shall be a combined total of 6 to 12 Design Reviews through Phase 1 and Phase 2.

C.3.068.3.1.2 [Orig] Design Reviews shall include the Preliminary Design Review (PDR), Detailed Design Reviews (DDRs), Critical Design Review (CDR), Production Readiness Review (PRR), Test Readiness Review (TRR), and Mockup Reviews. Reviews may be combined. The Post-Operational Test Design Review (Post-OTDR) is a Design Review but occurs after OP and is not considered part of Phase 1 or Phase 2.

C.3.068.3.1.3 [Orig] Design Reviews shall be held in person at the Contractor's facility or at an alternate location as agreed to by both parties and shall have a video teleconference for additional off-site USCG representation unless otherwise agreed to by both parties.

C.3.068.3.1.4 [Orig] The Contractor shall report on weight and centers of gravity trends, margins, and limits at each Design Review.

C.3.068.3.2 [Orig] The Preliminary Design Review (PDR)

C.3.068.3.2.1 [Orig] The Preliminary Design Review (PDR) shall be conducted when the Contractor understands the government contract design and has the processes and systems in place to enable detail design. The PDR shall be the first design review and marks the beginning of Phase 1: Design.

C.3.068.3.2.2 [Orig] The PDR shall include a USCG lead overview of the entire boat contract design intended for development as well as a briefing of the DDP.

C.3.068.3.3 [Orig] Detailed Design Reviews (DDRs)

C.3.068.3.3.1 [Orig] Detailed Design Reviews (DDRs) are Design Reviews conducted to concentrate on a specific topic, area, or function of the boat, and will be designated numerically (e.g., DDR-1, DDR-2, etc.).

C.3.068.3.3.2 [Orig] DDRs may be conducted in a manner that best communicates the relevant information and may be held in person at a facility provided by the Contractor, via computer conference, or a combination of the two as agreed to by both parties.

C.3.068.3.3.3 [Orig] Each DDR shall have approximately equal USCG review effort.

C.3.068.3.4 [Orig] Critical Design Review (CDR)

C.3.068.3.4.1 [Orig] The Critical Design Review (CDR) shall encompass a comprehensive review of the entire boat design as intended for construction and shall establish the Developmental Configuration Baseline (DCB).

C.3.068.3.4.2 [Orig] The CDR shall be held at the conclusion of Phase 1 to establish acceptance of all design/development deliverables.

C.3.068.3.4.3 [Orig] Outstanding design concerns from any DDR shall be resolved prior to completion of CDR and establishment of DCB.

C.3.068.3.5 [Orig] Production Readiness Reviews (PRRs)

C.3.068.3.5.1 [Orig] Production Readiness Reviews (PRRs) will assess the design maturity for transition to production of the first boat.

C.3.068.3.5.2 [Orig] Completion of a PRR is required prior to beginning production of the first boat. PRR's for subsequent hulls may consolidated and be requested by providing the Contractor or Contracting Officer/COR notice 30 calendar days in advance or as agreed to by the Contracting Officer.

C.3.068.3.5.3 [Orig] Limited release for production may be authorized by interim PRRs.

C.3.068.3.5.4 [Orig] A PRR may be identified to authorize purchase of LLTM.

C.3.068.3.6 [Orig] Mockup Reviews

C.3.068.3.6.1 [Orig] General. Mockup Reviews afford the USCG the opportunity to visualize and interact with the proposed configuration of certain spaces, check for contract compliance, and validate requirements prior to construction of the first boat.

C.3.068.3.6.2 [Orig] During a Mockup Review, the Contractor shall:

C.3.068.3.6.2.1 [Orig] Make provisions for and support the review effort;

C.3.068.3.6.2.2 [Orig] Make personnel available to resolve questions and to demonstrate equipment operation, maintenance accessibility, and removal and installation; and

C.3.068.3.6.2.3 [Orig] Collect consolidated feedback from USCG participants.

C.3.068.3.7 [Orig] Test Readiness Review (TRR)

C.3.068.3.7.1 [Orig] General. The Contractor shall conduct a Test Readiness Review (TRR) 60 days prior to starting One-Time Tests and Acceptance Testing.

C.3.068.3.8 [Orig] Post-Operational Test Design Review (Post-OTDR)

C.3.068.3.8.1 [Orig] General. The Contractor shall conduct the Post-OTDR 60 days following notification by the USCG that Operational Test has concluded and the USCG has delivered the OT report to the contractor, or as agreed to by both parties.

C.3.068.3.8.2 [Orig] Post-OTDR shall include an overview of all changes to be made to the HSC-L design as a result of OT that is agreed by both parties.

C.3.068.3.8.3 [Orig] the Post-OTDR may be conducted in a manner that best communicates the relevant information and may be held in person at a facility provided by the Contractor, via computer conference, or a combination of the two as agreed to by both parties.

C.3.068.4 [Orig] The Contractor shall establish and maintain a Systems Engineering Program that includes the core processes for managing and implementing the engineering effort and requirements management, to ensure through design, development, integration, test, production, training and support activities, that the boat and crew form an integrated system that meets the requirements of the HSC-L System Specification.

C.3.068.5 [Orig] The Contractor shall obtain and provide certification letters from the OEM for diesel engines, gear box and crane, verifying that the equipment is installed on the boat in compliance with OEM requirements.

C.3.073 [Orig] Noise and Vibration

C.3.073.1 [Orig] General

C.3.073.1.1 [Orig] The Contractor shall commission/consult an acoustic engineering firm with experience in marine sound mitigation to review the HSC-L design. The acoustic engineering firm shall participate in the Contractor's design review process and provide independent recommendations related to sound mitigation measures to be incorporated into the HSC-L design.

C.3.073.1.2 [Orig] The Contractor shall perform and provide reports on noise and vibration analysis to include but not limited to Airborne Noise Control, Propulsion System Vibration, Structural Vibration, and Generator Vibration. **[CDRL]**

C.3.076 [Orig] Material Certifications

C.3.076.1 [Orig] General

C.3.076.2 [Orig] Material certifications shall be maintained by the Contractor and available upon request from the USCG.

C.3.077 [Orig] Production Safety and Accident Prevention

C.3.077.1 [Orig] General

C.3.077.1.1 [Orig] The Contractor shall inform USCG personnel of its accident prevention programs, policies, and procedures by submitting a copy of its accident prevention plan or equivalent. **[CDRL]**

C.3.077.2 [Orig] Hazardous Materials (HAZMAT)

C.3.077.2.1 [Orig] The Contractor shall provide a Safety Data Sheet (SDS) Report that contains safety data information for each hazardous material (HAZMAT) used in the HSC-L. The report shall include an index of all SDSs. **[CDRL]**

C.3.080 [Orig] Integrated Logistics Support (ILS)

C.3.080.1 [Orig] Definitions

C.3.080.1.1 [Orig] Diminishing Manufacturing Sources and Material Shortages (DMSMS). An issue concerning the loss, or impending loss, of manufacturers or suppliers of items, raw materials, or software.

C.3.080.1.2 [Orig] Statement of Prior Submission (SPS). An SPS is a certificate signed by Contractor attesting that materials, analyses, reports, and/or equipment used on a HSC-L are identical in every respect to the item previously approved on an earlier HSC-L.

C.3.080.2 [Orig] General

C.3.080.2.1 [Orig] The Contractor shall establish an Integrated Logistics Support (ILS) program to satisfy the logistics requirements of the Contract. ILS program data shall be integrated and consistent with the engineering and technical documentation.

C.3.080.3 [Orig] Detailed Requirements

C.3.080.3.1 Contractor shall provide and implement an Integrated Logistics Support Management Plan (ILSMP)

C.3.080.3.1.1 [Orig] The ILS delivery schedule [CDRL] shall be integrated into the IMS.

C.3.080.3.2 [Orig] Logistics Product Data (LPD) is the set of Data Items that shall be submitted on the same schedule and on a system-by-system basis. LPD is comprised of the following Data Items:

C.3.080.3.2.1 [Orig] Bill of Materials for Logistics and Supply Chain Risk Management **[CDRL]**

C.3.080.3.2.2 [Orig] Provisioning Technical Data **[CDRL]**

C.3.080.3.2.3 [Orig] Commercial-off-the-Shelf Manuals **[CDRL]**

C.3.080.3.2.4 [Orig] System Manuals **[CDRL]**

C.3.080.3.2.5 [Orig] The LPD submittals shall be labelled with the system or subsystem UOC, and LCN as specified in paragraph C.3.080.3.2.8.

C.3.080.3.2.6 [Orig] The Contractor shall develop, when possible, the LPD Data Items in the following priority:

C.3.080.3.2.6.1 [Orig] **Category A (Most Complex and Mission Critical Systems)** in the following SWBS:

C.3.080.3.2.6.1.1 [Orig] 233 Diesel Engines

C.3.080.3.2.6.1.2 [Orig] 241 Propulsion Reduction Gears

C.3.080.3.2.6.1.3 [Orig] 310 Ship Service Generators

C.3.080.3.2.6.1.4 [Orig] 589 Cranes

C.3.080.3.2.6.2 [Orig] **Category B (Complex Systems)** in the following SWBS:

C.3.080.3.2.6.2.1 [Orig] 202 Machinery Centralized Control

C.3.080.3.2.6.2.2 [Orig] 324 Switchboard

C.3.080.3.2.6.2.3 [Orig] 421 Non-Electronic Navigation System

C.3.080.3.2.6.2.4 [Orig] 441 Exterior Communications System

C.3.080.3.2.6.2.5 [Orig] 581 Anchor

C.3.080.3.2.6.3 [Orig] **Category C** in the following SWBS:

C.3.080.3.2.6.3.1 [Orig] 512 Heating, Ventilation and Air Conditioning

C.3.080.3.2.6.3.2 [Orig] 520 Seawater Service

C.3.080.3.2.6.3.3 [Orig] 529 Drainage and Ballasting Systems

C.3.080.3.2.6.3.4 [Orig] 533 Potable Water Systems

C.3.080.3.2.6.3.5 [Orig] 541 Fuel Systems

C.3.080.3.2.6.3.6 [Orig] 555 Fire Extinguishing System

C.3.080.3.2.6.3.7 [Orig] 556 Hydraulic Oil Systems

C.3.080.3.2.6.3.8 [Orig] 593 Environmental Pollution Control Systems (Sewage System)

C.3.080.3.2.6.4 [Orig] **Category D** (Other systems, subsystems, and equipment):

C.3.080.3.2.6.4.1 [Orig] 167 Hull Structural Closures

C.3.080.3.2.6.4.2 [Orig] 251 Combustion Air System

C.3.080.3.2.6.4.3 [Orig] 252 Propulsion Control Systems

C.3.080.3.2.6.4.4 [Orig] 303 Protective Devices for Electrical Circuits

C.3.080.3.2.6.4.5 [Orig] 314 Power Conversion Equipment

C.3.080.3.2.6.4.6 [Orig] 330 Lighting Systems

C.3.080.3.2.6.4.7 [Orig] 405 Antennas

C.3.080.3.2.6.4.8 [Orig] 423 Electronic Navigation Systems

C.3.080.3.2.6.4.9 [Orig] 436 Alarms

C.3.080.3.2.6.4.10 [Orig] 528 Plumbing and Deck Drains

C.3.080.3.2.7 [Orig] Data format. The Contractor shall provide LPD in accordance with SAE GEIA-STD-0007B.

C.3.080.3.2.7.1 [Orig] SAE GEIA-STD-0007B Guidance. In order to ensure the data is provided in a logical manner, the Contractor shall use the following schema to label all Logistics Product data with the following requirements:

C.3.080.3.2.7.1.1 [Orig] Useable On Code (UOC). Each HSC-L hull shall have its own UOC. The UOC shall be the last three digits of the hull number. For example, if the first hull is to be numbered “950”, then its UOC is “950”.

C.3.080.3.2.7.1.2 [Orig] LSA Control Number (LCN) and Alternate LCN. This value shall be the Hierarchical Structure Code (HSC) in accordance with **[CDRL]** Configuration Status Report.

C.3.080.3.3 [Orig] ILS Program Review meetings. The Contractor shall conduct joint ILS Program Review meetings as part of the Quarterly PMR.

C.3.080.3.3.1 [Orig] At a minimum, this meeting shall address actions which were completed since the last PMR, current issues, logistics requirements, and upcoming activities for the following topics:

C.3.080.3.3.1.1 [Orig] Maintenance Planning

C.3.080.3.3.1.2 [Orig] Configuration Status Accounting

C.3.080.3.3.1.3 [Orig] Supply Support

C.3.080.3.3.1.4 [Orig] Provisioning Technical Data

C.3.080.3.3.1.5 [Orig] Outfitting

C.3.080.3.3.1.6 [Orig] Technical Manuals and Other Data

C.3.080.3.3.2 [Orig] The Contractor shall solicit additional agenda items for the ILS Program reviews from the Government ILS representative.

C.3.080.3.3.3 [Orig] The shipbuilder shall have the following available for review at all ILS Program Reviews:

C.3.080.3.3.3.1 [Orig] Technical data for the specific equipment being provisioned shall be available for review.

C.3.080.3.3.3.2 [Orig] The latest issue of drawings relevant to the logistics topics being reviewed.

C.3.080.4 [Orig] Diminishing Manufacturing Sources and Material Shortages (DMSMS) Forecast Report. The Contractor shall provide and maintain a DMSMS Plan. In addition, the contractor shall provide periodic DMSMS Health Assessment Reports. **[CDRL]**

C.3.080.5 [Orig] Supply Chain Risk Management. The Contractor shall implement supply chain assurance to reduce risk of counterfeit and non-conforming supplies and ensure resilience of the supply chain. The Contractor shall source critical components only from original manufacturers or authorized suppliers and maintain documentation of authenticity for such.

C.3.080.6 [Orig] The Contractor shall provide a Bill of Materials for Logistic and Supply Chain Risk Management incrementally by system in accordance with the approved ILS delivery schedule.**[CDRL]**

C.3.080.7 [Orig] Interim Contractor Logistics (ICLS) Requirements

C.3.080.7.1 [Orig] The Contractor shall provide non-warranty technical support, material, and labor.

C.3.080.7.2 [Orig] Technical support may include analysis, studies, inspections, surveys, parts, on-site casualty response, maintenance, upgrades, software updates, or documentation.

C.3.080.7.3 [Orig] Material may include any item or equipment installed on the boat.

C.3.080.7.4 [Orig] Labor may include a need for technical representatives, including Original Equipment Manufacturer (OEM) support and any required travel.

C.3.080.8 [Orig] ICLS During Operational Testing (OT).

C.3.080.8.1 [Orig] The HSC-L Operational Testing (OT) period is expected to be six months in duration and operate out of one of the USCG Stations located on the USCG District Northeast or District East.

C.3.080.8.2 [Orig] The Contractor shall provide these services for each month that this CLIN is ordered during the OT period:

C.3.080.8.2.1 [Orig] A dedicated technician shall be immediately available on-site Monday through Friday, 0800 – 1600 (US Eastern time) to assist OT boat operators to perform any required minor maintenance, repair, and corrective actions in order to minimize operational breaks.

C.3.080.8.2.2 [Orig] Phone technical support shall be available to USCG boat operators 24 hours a day, Monday through Friday (excluding Federal Holidays).

C.3.080.8.2.3 [Orig] Casualty response shall be available within 24 hours of notification to the Contractor of an operational failure beyond the capability of the on-site personnel to correct.

C.3.081 [Orig] Maintenance Planning

C.3.081.1 [Orig] General

C.3.081.1.1 [Orig] Maintenance planning shall be integrated with the HSC-L design, reliability and maintainability, manpower optimization, human performance factors, material selection, built-in test equipment, and other logistics processes. The HSC-L shall be designed and constructed to minimize maintenance requirements. This constraint shall not supersede any other specific performance, design, construction, or warranty requirements.

C.3.081.2 [Orig] Detailed Requirements

C.3.081.2.1 [Orig] For any equipment requiring periodic calibration or measurement, the Contractor shall ensure that at time of delivery no less than six months remains prior to the next required calibration or measurement.

C.3.083 [Orig] Supply Support

C.3.083.1 [Orig] General

C.3.083.1.1 [Orig] The Contractor shall make all HSC-L parts and components available for purchase.

C.3.083.2 [Orig] Master Equipment Configuration List (MECL)

C.3.083.2.1 [Orig] The Contractor shall develop a Master Equipment Configuration List (MECL) using the Government provided Major Equipment List (Attachment XXX), that represents, in a relational data structure, the configuration of the boat. The MECL shall be developed as a hierarchical configuration tree containing all Configuration Items (CI) that are part of the boat. The MECL shall be prepared electronically in a spreadsheet compatible with Microsoft Excel using the MECL spreadsheet template provided as GFI.

C.3.083.2.2 [Orig] Each CI listed on the MECL shall be provided with SOW Section C.3.083.7. provisioning data.

C.3.083.2.3 [Orig] Each CI listed on the MECL shall contain at a minimum accurate data in the following fields:

C.3.083.2.3.1 [Orig] ESWBS Number – The spreadsheet shall be sorted by the ESWBS, smallest to largest.

C.3.083.2.3.2 [Orig] Provisioning Document Control Number (PDCN) – The format for the number is made up of two letters and four numbers. For the HSC-L MECL, the PDCN shall start with "HL0001" for the first item in the spreadsheet followed by the next sequential number "HL0002." The PDCN shall be inserted after the MECL has been sorted in accordance with ESWBS smallest to largest.

C.3.083.2.3.3 [Orig] Configuration Item Functional Description (CIFD) – This shall be a functional description of the CI. Items may be grouped together to reduce the number of CIs, indicated by including the word "Group."

C.3.083.2.3.4 [Orig] Quantity – The number of components/equipment to be installed per boat.

C.3.083.2.3.5 [Orig] Unit of Issue – The unit of issue which pertains to the quantity reported.

C.3.083.2.3.6 [Orig] Price – To the nearest dollar in US currency.

C.3.083.2.3.7 [Orig] Contractor Commercial and Government Entity (CAGE) Code.

C.3.083.2.3.8 [Orig] Original Equipment Manufacturer (OEM).

C.3.083.2.3.9 [Orig] OEM Part Number.

C.3.083.2.3.10 [Orig] Firmware Version.

C.3.083.2.3.11 [Orig] National Stock Number (NSN) – The Contractor shall consult the Federal Logistics Information System (WebFLIS) for items on the MECL to check for existing National Stock Numbers (NSN) and include these NSNs on the MECL if applicable.

C.3.083.2.3.12 [Orig] Warranty Duration – This field shall be filled with the contract warranty duration for the CI, unless the OEM has a warranty for its product for a longer period. In that case, the longer duration shall be reflected within the MECL.

C.3.083.2.3.13 [Orig] Technical Manuals Source – This field shall indicate the technical publication source of the CI or "NA," indicating no technical publication source is associated with the CI.

C.3.083.2.3.14 [Orig] Preventive Maintenance Required – This field shall be either "Y," indicating preventative maintenance is required, or "N," indicating preventive maintenance is not required.

C.3.083.2.3.15 [Orig] Day Lead Time – "X" indicates the equipment has a commercial lead-time of over 30 days.

C.3.083.2.3.16 [Orig] Line-Item Deliverable – This column shall be marked with an "X," indicating that the Contractor would recommend that this CI be available within the USCG logistics system as a spare part for the boat.

C.3.083.2.3.17 [Orig] Serialized (Y/N) – This field shall be either "Y," indicating that this CI is listed in the Serial Number Report (SNR), or "N," indicating that the CI is not listed in the SNR.

C.3.083.2.3.18 [Orig] HSC-L Source Drawing – The product drawing that has this CI identified in the drawing Material List. The information in the product drawing shall match the corresponding MECL data.

C.3.083.2.3.19 [Orig] Engineering Change Notice Number – This will only be required if the CI is affected by an approved engineering change.

C.3.083.2.3.20 [Orig] The Contractor shall add additional data fields such as Purchase Order (PO) number, PO line item, PO date activated, and amplifying information, as required. This field shall indicate the PO number submitted under SOW Section 083-7.1.

C.3.083.3 [Orig] Serial Number Report (SNR)

C.3.083.3.1 [Orig] The Contractor shall provide a Serial Number Report (SNR) for each boat. Equipment on SNR shall be the serialized equipment found on the MECL. **[CDRL]**

C.3.083.4 [Orig] General Use Consumable List (GUCL). This is a list of recommended non-equipment related consumables necessary to support a boat's routine and administrative operations. **[CDRL]**

C.3.083.5 [Orig] List of Special Tools and Test Equipment

C.3.083.5.1 [Orig] The Contractor shall develop a list of special tools, test equipment, and software needed for maintenance and provide that list to the USCG**.[CDRL]**

C.3.083.5.2 [Orig] The list shall identify all special tools, the source of supply, and any special test equipment that would be required for maintenance and troubleshooting of the boat.

C.3.083.5.3 [Orig] A special tool is any tool or software designed and sold by the Contractor or a component OEM with unique characteristics for a specific maintenance task on a specific piece of equipment that cannot be accomplished with standard hand tools.

C.3.083.5.4 [Orig] If a unique connector or tool is required to support electronic diagnostics, the connector or tool shall be listed as a special tool (e.g., dongle).

C.3.083.5.5 [Orig] If unique software is required, the version number or other identifying information shall be provided.

C.3.083.6 [Orig] Operating Space Item (OSI). Support items required to perform routine tasks and stored near or in operating spaces. OSI may consist of equipage, loose hardware, consumables, accessories, tools, and support and test equipment. In certain cases, Storeroom Items (SRI) may become OSI due to the size or weight of the item. These are referred to as "bulkhead mounted spares".

C.3.083.7 [Orig] Provisioning Data Requirements

C.3.083.7.1 [Orig] Provisioning data, including copies of purchase orders and purchase order specifications, shall be provided for each end item, component or assembly and all support items which can be disassembled, reassembled, or replaced, and which, when combined, constitutes the end item, component, or assembly. Provisioning Data should provide the CG the ability to locate and purchase repair/replacement parts. **[CDRL]**

C.3.083.7.1.1 [Orig] OEM cut sheets indicating exact and complete part numbers are acceptable.

C.3.083.7.1.2 [Orig] Provisioning data shall include matching information for the OEM with CAGE code, OEM part number, and OEM pricing for end item, component and/or assembly for all listed on the MECL, at a minimum.

C.3.083.7.1.3 [Orig] Provisioning data shall include all pertinent information covered in SOW Section 086: Technical Manuals and Publications.

C.3.083.7.2 [Orig] For equipment, end items and components with embedded software or firmware, such as engine control management systems, propulsion control systems, and systems employing programmable logic controllers (PLCs), the Contractor shall ensure provisioning data information includes the version or other identifying information of the embedded software or firmware. For all software and firmware, the Contractor shall ensure the USCG has data rights necessary for the USCG to operate, maintain, diagnose, and repair the HSC-L system throughout its lifecycle.

C.3.083.7.3 [Orig] For any item on the MECL with a CAGE for the Contractor without a National Stock Number (NSN), additional Provisioning Technical Data Requirements shall be provided, including design drawings, manufacturing instructions, and additional technical data sufficient for the government to manufacture the piece of equipment.

C.3.084 [Orig] Preservation, Packing and Marking

C.3.084.1 [Orig] Shipments. The Contractor shall utilize ASTM D3951, for shipments that are:

C.3.084.1.1 [Orig] Items intended for immediate use.

C.3.084.1.2 [Orig] Small parcel shipments within Continental United States (CONUS), not-for-stock.

C.3.084.1.3 [Orig] Direct Vendor Deliveries (DVD) (CONUS ONLY).

C.3.084.1.4 [Orig] Items for boat outfitting.

C.3.084.2 [Orig] Storeroom Items. Packaging shall prevent damage caused by movement or weight of parts in a box. A Packing List itemizing parts in the box shall be mounted inside the lid of each box.

C.3.084.2.1 [Orig] The Contractor shall inspect and repackage (if needed) spare parts for binning and storage. Marking shall be in accordance with MIL-STD-129 and bar coding shall be in accordance with the ISO/IEC-16388-2007 Code 39 Symbology.

C.3.084.2.2 [Orig] Items removed from containers and placed in racks or shelf type stowage shall be protected against physical and environmental damage during storage.

C.3.084.3 [Orig] System Stock and Insurance Spares. Items designated for system stock and insurance spare shipments shall be packed in accordance with MIL-STD-2073-1 and MIL-DTL-2845, labelled in accordance with MIL-STD-129, and with bar coding in accordance with ISO/IEC-16488-2007 Code 39 Symbology.

C.3.084.4 [Orig] Packing and Preservation Data. The Contractor shall develop Packing and Preservation data and, when required, Special Packing Instructions (SPI) for system stock and insurance spares in accordance with MIL-STD-2073-1 and document the results of the analyses in the Preservation and Packing Data Report. **[CDRL]**

C.3.084.5 [Orig] Hazardous Materials. For items containing HAZMAT that cannot be packaged in accordance with MIL-STD-2073-1, the contractor shall perform and shall acquire SDS data necessary on the hazardous material, to support compliance with the Performance Oriented Packaging (POP) requirements of HAZMAT as defined in Title 49 CFR (Motor), the International Maritime Organization's International Maritime Dangerous Goods Code (IMDG) (Vessel), the International Civil Aviation Organization (ICAO) (Commercial Air), Air Force Manual 24-204 (AFMAN) (Military Air), and Defense Transportation Regulation, Part II, Cargo Movement (DOD 4500.9R) Technical Instructions for Safe Transport of Hazardous Goods.

C.3.084.6 [Orig] Containerization. For those items that require containerization per MIL-STD-2073-1 and MIL-DTL-2845, the Contractor shall request a search of the Department of Defense (DoD) Container Design Retrieval System (CDRS) for any specialized reusable container designs. **[CDRL]**

C.3.084.6.1 [Orig] If the CDRS system has a suitable container design, the Contractor shall use that for the subject item. If CDRS does not have a suitable container, the Contractor shall design and fabricate a new container in accordance with SAE ARP1967B and provide the data for input to CDRS. **[CDRL]**

C.3.084.6.2 [Orig] The Contractor shall fabricate the container and package items with the container for shipment to the Government designated location.

C.3.085 [Orig] Drawings

C.3.085.1 [Orig] General Drawing Requirements

C.3.085.1.1 [Orig] The Contractor shall prepare and format drawings in AutoCAD compatible (.dwg) file format and in accordance with COMDTINST M9085.1, except as modified herein. External reference files (XREF) shall be bound or inserted into drawing files. (AutoCAD revision is recommended to be 2017 or later.)

C.3.085.1.1.1 [Orig] The Contractor shall only use Standard USCG drawing templates A, B, C, D, H-8 or H-12.

C.3.085.1.1.2 [Orig] Each drawing shall be assigned a standard USCG drawing number.

C.3.085.1.1.2.1 [Orig] Asset Class designation used in drawing numbers will be assigned by the Government.

C.3.085.1.1.3 [Orig] Electronic file naming convention for construction drawings shall comply with COMDTINST M9085.1 Chapter 4.

C.3.085.1.1.4 [Orig] The actual physical components depicted (drawing geometry) shall be at full scale (1:1) in “Model Space” with scaled viewports utilized in “Paper Space”.

C.3.085.1.1.5 [Orig] Text and dimensions in body of drawing shall use romans.shx font with a minimum height of 0.10" and maximum width factor of 0.8. Text and dimensions shall be in "Paper Space" except for embedded text in block geometry.

C.3.085.1.1.6 [Orig] Drawing view callout text shall be placed underneath the middle of all drawing views on each drawing sheet. View zone number and letter for the view title shall come from where the center of the callout text is located in the body of the drawing under each drawing view (e.g. “PLAN VIEW 5-A” has the callout text located in horizontal zone “5” and vertical zone “A”). This pertains to all views, even if the view expands into multiple zones of the drawing sheet. A Callout dynamic block with required text height, view titles and instructions for usage is furnished on each sheet of each GFI drawing.

C.3.085.2 [Orig] Drawing Number Assignment Report. The Contractor shall provide and maintain a Drawing Number Assignment Report. **[CDRL]** The Contractor shall assign a drawing number to each design product developed for the HSC-L.

C.3.085.3 [Orig] Referenced Drawings. The Contractor shall provide drawings referenced within any drawing that have not been previously delivered under this contract.

C.3.085.4 [Orig] Drawing Conventions. The Contractor shall utilize the following drawing conventions:

C.3.085.4.1 [Orig] Identify the first revision of Construction Drawings as “FIRST SUBMITTAL”. If the drawing is required to be re-submitted, it shall have the next sequential numerical identifier shown and in the revision block the revisions made corresponding to the comments received from the Government.

C.3.085.4.2 [Orig] The as-built drawings shall have a "-" in the revision area in the drawing number blocks throughout the drawings. If the drawing is required to be re-submitted, it shall have the next sequential alphabetical identifier.

C.3.085.4.3 [Orig] Label plate and signage shown on drawings shall incorporate the format and labeling requirements defined in the HSC-L System Specification.

C.3.085.4.4 [Orig] Drawings shall include hull applicability.

C.3.085.4.5 [Orig] Views from vendor drawings of equipment may be used on the drawing with reference to the vendor drawing for complete information.

C.3.085.4.6 [Orig] Material List. For arrangement and detail drawings, physical components within the drawing shall be listed on a single Material List.

C.3.085.4.6.1 [Orig] Physical components include bulk materials such as adhesives, except that welding consumables need not be listed.

C.3.085.4.6.2 [Orig] The Material List shall be standardized.

C.3.085.4.6.3 [Orig] Items on the Materials Lists shall include their ESWBS-based HSC as assigned in **[CDRL]** Configuration Status Reports.

C.3.085.4.6.4 [Orig] The Material List shall include: Material Specifications, Hardware, and Quantities.

C.3.085.4.6.5 [Orig] For each component, provide the salient characteristics needed to re-procure the item.

C.3.085.4.6.6 [Orig] Whenever available, include the OEM and their part number for the component.

C.3.085.4.6.7 [Orig] Distributors' part numbers may be included in addition to the salient characteristics, but are not an acceptable substitute for those salient characteristics or the OEM part number.

C.3.085.4.6.8 [Orig] Whenever distributors' part numbers are used, the entry shall include the words "or equal" to differentiate it from original equipment manufacturer information as well as provide the salient characteristics needed to reproduce the item in a competitive solicitation.

C.3.085.4.6.9 [Orig] NSNs shall be included for components if available.

C.3.085.4.6.10 [Orig] The item numbers from the parts list or bill of material shall be shown in the field of the drawing with leaders to the various components in the included views.

C.3.085.5 [Orig] Boat Construction Drawings. Two-Dimensional (2D) drawings shall be developed to demonstrate that the boat conforms to the requirements of the HSC-L System Specification and include diagrams depicting functional interconnections between components, arrangement drawings and associated lists. **[CDRL].**

C.3.085.6 [Orig] System Arrangements Drawings. Components including piping, electrical wiring and other components, except minor components such as cable ties, adhesives or other bulk materials on the boat, shall be depicted on an arrangement drawing in their true size, location and orientation. Purchased components shall be depicted as simple shapes having the same outside dimensions of the object depicted. Interfaces created by the Contractor shall be depicted including fasteners, pipe joints and terminal strips.

C.3.085.6.1 [Orig] Non-standard items not purchased shall have fabrication sketches provided.

C.3.085.7 [Orig] 3D Technical Data Package. The Contractor shall provide the Production Level 3D Technical Data Package (TDP) per MIL-STD-31000, consisting of a 3D Product Model, Construction drawings, design, manufacturing and technical data, and associated lists. The 3D TDP shall support operational use, logistics support, repair, maintenance, and construction/manufacture of the HSC-L. The 3D TDP shall include the following minimum functionality and attributes: **[CDRL]**

C.3.085.7.1 [Orig] The 3D Product Model shall be maintained to the approved current product baseline.

C.3.085.7.2 [Orig] The 3D Product Model shall be provided both in its native format and an ISO 10303 STEP compliant format.

C.3.085.7.3 [Orig] 2D Drawings extracted from the 3D Product Model shall be fully compatible with AutoCADTM without the use of any add-on or viewer programs.

C.3.085.7.4 [Orig] Physical components shall be depicted or modeled in their true location, shape and orientation in one or more 3D CAD files.

C.3.085.7.5 [Orig] The 3D Product Model shall contain one or more linked 3D CAD files and associated electronic lists. The native format 3D Product Model shall be developed in accordance with MIL-STD-31000.

C.3.085.7.6 [Orig] The 3D Product Model shall not have any redundant models of parts, components, or geometry. There shall be only one model of any physical item in the 3D Product Model.

C.3.085.7.7 [Orig] The 3D Product Model may contain depictions of components in an in-process condition, such as flat parts prior to bending or forming, cast parts prior to machining, and similar parts in a form not ready to install on the boat, at the Contractor's option. This shall not be considered a redundant depiction.

C.3.085.7.7.1 [Orig] Any depictions or models of parts in an in-process condition shall be readily distinguished from the as-installed depictions by layering conventions, distinctive part numbers, separable externally referenced files or similar techniques that allow such in-process parts to be readily removed from a display or other database.

C.3.085.7.7.2 [Orig] In-process parts that do not have a meaningful location prior to final placement in the model shall be located as close to the expected location as possible.

C.3.085.7.7.3 [Orig] In-process parts may also be omitted from the 3D Product Model.

C.3.085.7.8 [Orig] A part may be depicted as a block, surface or solid showing the volume of space it occupies to a precision appropriate to the need for avoiding interferences with the installed component. A small part may be designated as a simple symbol in the Contractor's format. As an example, a fastener may be depicted as a block comprising a line with attached attributes describing it, but shall not be depicted as a solid with threads modeled.

C.3.085.7.9 [Orig] The 3D Product Model shall depict the Space and Weight envelopes defined in the HSC-L System Specification.

C.3.085.7.10 [Orig] Any component, including structural components that are subsequently joined by welding or any other means, shall include as a separable attribute the weight and the x, y, and z coordinates of the center of gravity. This information shall be a linked, non-graphic entity capable of being extracted from the 3D Product Model by automatic processes.

C.3.085.7.11 [Orig] The Contractor shall maintain the Production Level 3D TDP to reflect each hull applicability.

C.3.085.7.12 [Orig] The Contractor shall deliver HW and software licenses, as well as provide technical support to the government for 10 standalone laptops to allow review of the 3D Product Model.

C.3.085.7.12.1 [Orig] The Contractor shall provide the government with up to 10 virtual/web-based user level training sessions on the 3D Product Model Software.

C.3.085.8 [Orig] System Diagrams. System diagrams shall depict the interconnections of the components in the system within the plan views of the boat structure. Components that provide functional effect on the system shall be depicted.

C.3.085.8.1 [Orig] Piping and Mechanical Diagrams shall include components, such as valves and piping connections, which can be disconnected in the course of normal servicing. Permanent joints such as threaded, welded or soldered pipe connections need not be depicted. Components and their interconnections shall be located as required for clarity in comprehending the functionality of the system and shall not be to scale in the diagram. **[CDRL]**

C.3.085.8.2 [Orig] Components shall be depicted as symbols and diagrams shall include a list of symbols used.

C.3.085.8.2.1 [Orig] Symbols for piping system drawings shall be in accordance with ASTM F1000, except hydraulic system drawings, which shall be in accordance with SAE J1780, using graphic symbols from ANSI Y32.10 as supplemented by SAE AS1290.

C.3.085.8.2.2 [Orig] Symbols for HVAC system drawings shall be in accordance with ASTM F856.

C.3.085.8.2.3 [Orig] Hose and pipe shall be distinguished from each other.

C.3.085.8.3 [Orig] Diagrams shall include tables with major component characteristics such as material types and specifications, flow rates, pressure settings, sizing information, etc. Where appropriate, flow quantities, pressures, directions and nominal pipe sizes shall be shown as text near or leader to the line depicted.

C.3.085.9 [Orig] Machinery Arrangement Drawings.

C.3.085.9.1 [Orig] Machinery arrangement drawings shall be provided showing the major components in the main and auxiliary spaces, in detail, to justify the basic configurations and space allocations reflected in the design. **[CDRL]**

C.3.085.10 [Orig] Electrical and Electronic Drawings.

C.3.085.10.1 [Orig] Graphic symbols for electrical and electronics diagrams shall be in accordance with IEEE-STD-315 or MIL-HDBK-290 and shall indicate type of equipment and fixtures.

C.3.085.10.2 [Orig] Schematic diagrams shall depict, by means of graphic symbols, connections and functions of circuit arrangements. The schematic diagram shall trace the circuit and its functions without regard to the actual physical size, shape, or location of the component devices or parts.

C.3.085.10.3 [Orig] Isometric System drawings shall show the system in isometric format and shall include information relative to vendor make and model number, cable types and sizes, material list, deck and bulkhead penetrations, location of hardware on the boat, referenced to the technical manuals, where applicable.

C.3.085.10.3.1 [Orig] The drawings shall depict point-to-point wiring connections to trace power, communication and control signals. Individual wires shall be terminated at identified terminals within equipment, including terminal boxes. Power supply source and signal destination shall be indicated.

C.3.085.10.3.2 [Orig] Wires requiring shielding and grounding shall be designated, and quantity of spare conductors shall be identified for each cable.

C.3.085.10.4 [Orig] Elementary wiring diagrams shall show the system in diagrammatic format and include information relative to vendor make and model number, cable types and sizes, material list, location of hardware on the boat, referenced to the technical manuals, where applicable.

C.3.085.10.4.1 [Orig] The diagrams shall depict point-to-point wiring connections to trace power, communication and control signals. Individual wires shall be terminated at identified terminals within equipment, including terminal boxes. Power supply source and signal destination shall be indicated.

C.3.085.11 [Orig] As-Built Drawings. The Contractor shall provide the complete set of As-Built Drawings for the lead boat and provide updated As-Built drawings for each follow boat. As-Built Drawings shall reflect the as-built, as-delivered configuration of the boat and shall be complete, without attached change paper or liens and with current revision blocks and numbers. **[CDRL]**

C.3.086 [Orig] Technical Manuals and Other Data

C.3.086.1 [Orig] Technical Manual Organization Plan. The Contractor shall provide and implement a Technical Manual Organization Plan (TMOP) **[CDRL]**

C.3.086.2 [Orig] Technical Data Status Report. The Contractor shall provide a Technical Data Status Report (TDSR) that provides a complete index and status of technical manuals. The report shall provide the status of development, validation, and planned delivery dates. **[CDRL]**

C.3.086.3 [Orig] Technical Manual Requirements. The Contractor shall provide technical manuals in accordance with MIL-DTL-24784 for installed equipment, components, systems, and for special purpose tools or test equipment.

C.3.086.3.1 [Orig] Deliver technical manuals to the Government in an unlocked, Portable Document Format (PDF).

C.3.086.3.2 [Orig] Assign a unique number to all technical manuals from a range provided by the Government.

C.3.086.3.3 [Orig] Include the following front matter in each technical manual in accordance with MIL-DTL-24784/4:

C.3.086.3.3.1 [Orig] Cover Page

C.3.086.3.3.2 [Orig] Data Rights Release Letter:

C.3.086.3.3.3 [Orig] Validation Certificate:

C.3.086.3.3.4 [Orig] Manual Evaluation Checklist

C.3.086.3.4 [Orig] Operation and Service Manuals for diesel engines shall include:

C.3.086.3.4.1 [Orig] Installation drawings.

C.3.086.3.4.2 [Orig] Engine sectional drawings.

C.3.086.3.4.3 [Orig] Table of normal clearances and measurements for new parts.

C.3.086.3.4.4 [Orig] Maximum allowable clearances, tolerances, measurements and wear limits for wearing parts critical to engine performance.

C.3.086.3.4.5 [Orig] Comparison photographs if used in lieu of measurements.

C.3.086.3.4.6 [Orig] Linear vibration limits and locations on the engine.

C.3.086.3.4.7 [Orig] Lube oil and jacket water analyses along with trending limits.

C.3.086.3.4.8 [Orig] Torque settings and hydraulic pressure of important threaded hardware.

C.3.086.3.4.9 [Orig] Information describing shipboard level maintenance including periodic and corrective maintenance.

C.3.086.3.5 [Orig] Commercial Off-The-Shelf (COTS) Manuals. COTS manuals shall be provided in accordance with MIL-DTL-24784/4. **[CDRL]**

C.3.086.3.5.1 [Orig] Supplemental data shall be developed when COTS manuals do not meet the content requirements defined in MIL-DTL-24784/4.

C.3.086.3.5.2 [Orig] Supplemental data shall be attached as an appendix to associated COTS manual.

C.3.086.3.5.3 [Orig] Publications prepared for this contract by the Contractor shall be in an electronic format that allows imbedded text and graphics and shall be in Microsoft® Office Word format or Microsoft® Office Excel format that is compatible with Windows® 10 and 11 or greater.

C.3.086.3.5.4 [Orig] Any Commercial Off-the-Shelf (COTS) manuals from vendors that are not in the Microsoft® Office Word or Excel compatible format shall be delivered in an unlocked PDF file.

C.3.086.3.5.5 [Orig] The Contractor shall provide an index of all technical manuals and publications for all equipment, components, and accessories that make up the boat.**[CDRL]**

C.3.086.3.5.6 [Orig] The Contractor shall provide an electronic copy of all technical manuals and publications for all equipment, components, and accessories that make up the boat.

C.3.086.3.6 [Orig] Data Rights: Data within the technical manuals and publications developed at private expense that embody trade secrets or are commercial or financial and confidential or privileged shall be delivered with limited rights and shall be treated in accordance with FAR Clause 52.227-14 (Alternatives II, IV, and V). All other data shall be delivered with unlimited rights.

C.3.086.3.6.1 [Orig] When it is necessary to indicate limited rights in accordance with FAR Clause 52.227-14 (Alternative II), it shall be clearly identified and the following statement shall be included on the first page of each applicable technical manual or publication:

> **"RIGHTS IN DATA**
>
> (a) These data are submitted with limited rights under Government Contract No. [To be Inserted at Contract Award]. These data may be reproduced and used by the Government with the express limitation that they will not, without written permission of the Contractor, be used for purposes of manufacture nor disclosed outside the Government; except that the Government may disclose these data outside the Government for the following purposes, if any; provided that the Government makes such disclosure subject to prohibition against further use and disclosure:

> (1) Use (except for manufacture) by support service Contractors; (2) Use (except for manufacture) by other Contractors participating in the Coast Guard's program of which this specific contract is a part; (3) Repair or overhaul work; and (4) Release to a foreign Government, or its instrumentalities, if required to serve the interests of the U.S. Government, such as in the context of the Foreign Military Sales program, for information or evaluation, or for emergency repair or overhaul work by the foreign Government. See FAR § 27.404-2(c)(i).
>
> (b) This notice shall be marked on any reproduction of these data, in whole or in part."

C.3.086.3.6.2 [Orig] Copyrights and proprietary information credit line; technical manuals shall not contain copyrighted material except as specified in the FAR and DHS Acquisition Regulations. When copyrighted material is to be included in a technical manual, the developer shall obtain prior written permission from the copyright owner or authorized agent for its use by the USCG. The signed, written permission shall be delivered together with the final technical manuals in accordance with the contract. The written permission shall contain a statement declaring whether or not a copyright credit line is required. When it is necessary to include copyright and proprietary material, it shall be clearly identified and the following warning statement shall be included on the title page:

> "This document contains copyright or proprietary materials. Infringement of copyright or proprietary material may violate existing Federal laws and statutes and result in criminal penalties, imprisonment, or removal from office."

C.3.086.3.7 [Orig] Technical manuals and publications shall include on the cover sheet/page the associated ESWBS number and sequential/serialized number if there are multiple submissions with the same ESWBS number.

C.3.086.4 [Orig] OEM Technical Manuals **[CDRL]**

C.3.086.4.1 [Orig] The Contractor shall deliver OEM technical manuals. These manuals shall be provided for equipment, components, and accessories that make up the boat.

C.3.086.4.2 [Orig] OEM documentation shall be in English.

C.3.086.4.3 [Orig] The Contractor shall provide all OEM technical manuals offered by an OEM. OEM technical manuals would typically include the necessary information and instructions to: install, operate, and maintain the equipment. This includes a description of system start-up, operation, shutdown, safety, and emergency procedures. The manuals would also contain information concerning inspection, preventive and corrective maintenance, calibration, adjustment, troubleshooting, illustrated parts breakdowns that include OEM part numbers, repair, and required tools and test equipment.

C.3.086.4.4 [Orig] If an OEM technical manual contains information on more than one like component, the Contractor shall mark each manual with the component model and part number used on the boat to direct the user to only those sections necessary to complete their task.

C.3.086.4.5 [Orig] The Contractor shall provide supplemental information not addressed by the OEM technical manual that covers the as-installed details and instructions for the OEM component used for the construction of the boat.

C.3.086.5 [Orig] Technical Publications

C.3.086.5.1 [Orig] The Boat Information Book (BIB)**[CDRL]** shall provide a complete description of the boat's capacities and characteristics, including operating instructions for all systems, equipment, and components installed (to include wiring). System diagrams shall be included to supplement the written description. System working drawings may be used in lieu of the diagrams upon approval from the USCG. ABYC T-24 Owner/Operator's Manuals shall be used as guidance when developing the content of the BIB. The BIB at a minimum shall contain the following sections:

C.3.086.5.2 [Orig] The Contractor shall develop a Hull, Mechanical & Electrical (HM&E) Support Section that provides the following:

C.3.086.5.2.1 [Orig] Physical layout diagrams and descriptions.

C.3.086.5.2.2 [Orig] A list of all embedded electronic control systems within the HM&E boundary.

C.3.086.5.2.3 [Orig] Step-by-step flow chart describing each equipment setup and startup processes and settings.

C.3.086.5.2.4 [Orig] System corrective maintenance, casualty recovery, and reset procedures.

C.3.086.5.2.5 [Orig] The Contractor shall develop a Command, Control, Communications, Computer, Cyber, Intelligence, Surveillance, and Reconnaissance (C5ISR) System/Network Support Section that provides the following:

C.3.086.5.2.5.1 [Orig] System overview including a general description of the nature of the system as a whole and the purpose and description of each major sub-system.

C.3.086.5.2.5.2 [Orig] Physical layout diagrams and descriptions.

C.3.086.5.2.5.3 [Orig] System Operation Verification Test Development Data

C.3.086.5.2.5.4 [Orig] AIS-2 GPS antenna

C.3.086.5.2.5.4.1 [Orig] A: Antenna Distance, Bow to Antenna

C.3.086.5.2.5.4.2 [Orig] B: Antenna Distance, Stern to Antenna

C.3.086.5.2.5.4.3 [Orig] C: Antenna Distance, Port Beam to Antenna

C.3.086.5.2.5.4.4 [Orig] D: Antenna Distance, Starboard Beam to Antenna

C.3.086.5.2.5.5 [Orig] SINS-2 GPS antenna

C.3.086.5.2.5.5.1 [Orig] A: Antenna Distance, Bow to Antenna

C.3.086.5.2.5.5.2 [Orig] B: Antenna Distance, Stern to Antenna

C.3.086.5.2.5.5.3 [Orig] C: Antenna Distance, Port Beam to Antenna

C.3.086.5.2.5.5.4 [Orig] D: Antenna Distance, Starboard Beam to Antenna

C.3.086.5.2.6 [Orig] Static draught for the depth sounder

C.3.086.5.2.7 [Orig] A list of all C5ISR equipment broken out into navigation and communication (external and internal) functional groups.

C.3.086.5.2.8 [Orig] Step-by-step flow chart describing each equipment setup and startup processes and settings.

C.3.086.5.2.9 [Orig] System corrective maintenance, casualty recovery, and reset procedures.

C.3.086.5.3 [Orig] Casualty Control Section - The Contractor shall develop a casualty control section that provides description of the emergency systems and action to be taken for the following:

C.3.086.5.3.1 [Orig] Capsizing

C.3.086.5.3.2 [Orig] Charlie Fire

C.3.086.5.3.3 [Orig] Flooding

C.3.086.5.3.4 [Orig] Engine High Water Temperature

C.3.086.5.3.5 [Orig] Fire in the Machinery Compartment(s)

C.3.086.5.3.6 [Orig] Loss of Control of Engine RPM

C.3.086.5.3.7 [Orig] Loss of Fuel Pressure

C.3.086.5.3.8 [Orig] Loss of Lube Oil Pressure

C.3.086.5.3.9 [Orig] No Power/Insufficient Power to Communications/Navigation Equipment

C.3.086.5.3.10 [Orig] Steering Casualty (Hydraulic/Electrical)

C.3.086.5.3.11 [Orig] Fouled Propeller

C.3.086.5.3.12 [Orig] Loss of Cooling Flow

C.3.087 [Orig] Facilities

C.3.087.1 [Orig] Office facilities and services for On-Site and visiting Government Representatives shall be located within the Contractor's boat construction facility(s) and/or any facility where a boat under this contract is berthed after construction (e.g., if the boat trials are in a different location from the construction yard). This may require duplicate office facilities when concurrent construction and trials of multiple boats under this contract occur at several locations.

C.3.087.2 [Orig] Office facilities shall have convenient access to the boat. Office facilities and services for Government Representatives shall be provided at each location where a boat under this contract is under construction or is berthed.

C.3.087.3 [Orig] Office facilities ***shall be dedicated (stand-alone) to the HSC-L program only***. Office facilities shall not be shared across other programs for the purpose of meeting other program contracted office requirements.

C.3.087.4 [Orig] Commencing four months after contract award through the delivery and acceptance of the last boat, the Contractor shall provide the requirements of this section exclusive of any service provided by/for any other contract:

C.3.087.4.1 <u>[Orig] Office Facility Requirements</u> – Standalone suitable office facilities and services shall be provided for the on-site Government staff of ten personnel which will include two private offices and one conference room. In addition, office facilities shall be provided for the crew (6 personnel/boat, maximum 24). Boat crews will arrive in the shipyard approximately six months prior to the boat transiting to the delivery location. The Contractor shall provide cleaning services at least weekly. Restrooms and kitchen facilities shall be available within vicinity of the Government office space. The Contractor shall be responsible for pest control.

C.3.087.4.2 <u>[Orig] Office Furniture and Equipment</u> – Appropriate office furniture and equipment shall include desk workstations with trash cans for all Government and crew.

C.3.087.4.3 [Orig] Conference Room – The Contractor shall provide access to one conference room with doors and tables with seating for at least twelve people. The conference room shall include a speaker phone, at least one screen, and teleconferencing capability.

C.3.087.4.4 [Orig] Telephone and Internet Service – The Contractor shall provide and maintain telephone with voicemail and high speed internet service for the on-site Government office spaces. Separate lines for internet service shall be provided to each workstation and three internet lines shall be provided for use in the Conference Room.

C.3.087.4.5 [Orig] Copier – The Contractor shall provide a multi-function color copier. The copier shall be capable of copying and printing up to 11x17" size paper, double-sided, and scanning.

C.3.087.4.6 [Orig] Document waste – The Contractor shall provide a high-quality paper shredder to collect business sensitive documentation. Recycle services shall be provided.

C.3.087.4.7 [Orig] Parking Spaces – The Contractor shall provide 10 designated, secure parking spaces located close to the office facility for the Government personnel.

C.3.087.4.8 [Orig] Access to the Boats(s) and Applicable Facilities – Government and authorized government representatives shall have admission to the plant, access to the vessel(s) where and as required, and be permitted, within the plant and on the vessel(s) to perform and fulfill their respective obligations to the Government.

C.3.088 [Orig] Human Systems Integration (HSI)

C.3.088.1 [Orig] The Contractor shall establish and implement a Human Systems Integration (HSI) program at every phase of design and testing in accordance with ASTM F1337 and planned in accordance with MIL-STD-46855A.

C.3.088.2 [Orig] HSI Risk and Issues Tracking. The Contractor shall identify risks and use the HSI Risk Analysis process, ASTM F1337, to assess their impact to human performance and mission success.

C.3.088.2.1 [Orig] The Contractor shall report on HSI efforts and top 10 identified risks with proposed mitigations at Milestone Events, PMRs, ILS Program Reviews, and Readiness Reviews.

C.3.088.2.2 [Orig] The Contractor shall correct design features that adversely impact human performance that are categorized as "high" or "serious" by the government in accordance with MIL-STD-882(series).

C.3.088.2.3 [Orig] The Contractor shall identify corrective actions or mitigations/workarounds needed due to system capability gaps.

C.3.088.3 [Orig] Human Factors Engineering.

C.3.088.3.1 [Orig] The Contractor shall establish human factors design requirements to develop and select Human-Machine Interfaces (HMIs) that are within the cognitive, physical, or sensory capabilities and limitations of the users; that do not require complex manpower or training intensive tasks; nor result in frequent or critical errors.

C.3.088.3.2 [Orig] The Contractor shall develop or select HMIs to produce a common look and feel with consistent protocols for system interfaces.

C.3.088.3.3 [Orig] The Contractor shall apply HFE in the selection and integration of Non-Developmental Items (NDI) into the design.

C.3.088.3.4 [Orig] 3D Product Model. During the design review process, the Contractor shall use the 3D Product Model Visualization to demonstrate usability (as defined in ISO 9241-11:2018) of the HSC-L and shall use it to demonstrate proposed configuration changes.

C.3.088.3.4.1 [Orig] The Product Model or Product Model Visualization shall be capable of demonstrating accessibility for operation, manipulation, removal or replacement tasks using an anthropometric model that simulates body dimensions that are design-critical to the task(s) from a 5th percentile female through a 95th percentile male, as defined in ASTM F1166, with required personal protective clothing and equipment and applicable task tools / peripherals.

C.3.088.3.4.2 [Orig] The 3D Product Model or Product Model Visualization shall provide the capability to support virtual walkthroughs for HSI verification by 5th percentile female through a 95th percentile male to allow visualization of the actions they need to take to perform their tasks and their sightlines for the following:

C.3.088.3.4.2.1 [Orig] Pilothouse operations (to include crane operations).

C.3.088.3.4.2.2 [Orig] Machinery Room(s) (maintenance activities).

C.3.088.3.4.2.3 [Orig] Towing, and mooring.

C.3.088.3.4.2.4 [Orig] Working Deck Operations.

C.3.088.3.4.3 [Orig] The Contractor shall demonstrate that the design supports human engineering requirements, mission requirements, and evaluate the arrangement and accessibility for operation and maintenance of equipment mounted or stowed at Design Reviews, Mockup Reviews, and PMRs.

C.3.092 [Orig] Test Administration and Testing

C.3.092.1 General

C.3.092.1.1 [Orig] The Contractor shall ensure that the boat construction has been completed and all systems have been groomed to verify they perform as intended before Acceptance Tests (ATs) are conducted.

C.3.092.1.2 [Orig] The Contractor shall develop an AT program to verify that the boat meets all the requirements of the contract and documents the results. The required tests are minimums and are not intended to supplant any controls, examinations, inspections, or tests normally employed by the Contractor to assure the quality of the boat. The boat shall successfully complete all testing prior to acceptance.

C.3.092.1.3 [Orig] The ATs detailed in this section are applicable to all boats unless stated otherwise.

C.3.092.1.4 [Orig] During all ATs, the boat shall remain under the control of the Contractor. The Contractor shall ensure that the boat and all equipment is operated in a safe manner in accordance with OEM-recommended practices and shall assume responsibility for all operations.

C.3.092.1.5 [Orig] The Contractor shall furnish all material, fuel, labor, power, equipment, instruments, and expertise necessary for these ATs.

C.3.092.1.6 [Orig] Any instruments used shall be calibrated and marked prior to use. The calibration shall be current. Instruments shall be calibrated using standards certified as being traceable to a national standard. Calibration data and certificates shall be available for inspection by USCG representatives. The Contractor shall ensure that each individual AT Report contains copies of calibration data and certificates for the equipment used in that test.

C.3.092.1.7 [Orig] At the satisfactory completion of all AT, the Contractor shall remove the testing instrumentation and equipment.

C.3.092.1.8 [Orig] When testing the equipment, components, or systems, all AT procedures shall be in accordance with component and equipment manufacturers' recommended practices and shall not violate component and equipment operating procedures or the manufacturer's warranty.

C.3.092.1.9 [Orig] The Contractor shall ensure proper operation of the equipment. Proper operation means that the equipment can be started, operated, interfaced together as required, and demonstrated to function in the designed manner. The Contractor shall conduct all tests regardless of whether the item to be tested is furnished by the USCG or the Contractor.

C.3.092.1.10 [Orig] The Contractor shall not test or operate the tactical VHF radio at power levels above 25 watts.

C.3.092.1.11 [Orig] If damage to any component or system occurs during or after any testing, and prior to acceptance of the boat, the damage shall be repaired by the Contractor and previously completed tests of the component or system damaged shall be repeated.

C.3.092.1.12 [Orig] Unless otherwise agreed to by the USCG, ATs shall be conducted from the contractor's facility.

C.3.092.1.13 [Orig] The USCG may, at its discretion, observe any AT. The Contractor may limit the number of USCG representatives onboard the boat during underway testing. Underway testing shall not last for more than eight hours without the opportunity to change out USCG observers.

C.3.092.1.14 [Orig] During ATs, the boat shall be operated in a manner and in waters suitable for collection of data for the accepted test procedure. Full power/endurance tests shall be conducted in water depths of at least 30 feet and allow for safe and prudent navigation of the boat under its own power at idle through full speed. The underway test shall be postponed or rescheduled if winds and seas are excessive, as determined by mutual agreement between the Contractor and the USCG representative.

C.3.092.1.15 [Orig] Underway tests shall be conducted in calm water condition and starting at the performance weight condition, unless otherwise specified. The boat may be ballasted temporarily to achieve the required loading condition.

C.3.092.2 [Orig] Administration and Conduct

C.3.092.2.1 [Orig] The Contractor shall notify the USCG in writing with the date, time, and ATs location. Notification shall be given during normal working hours (Monday – Friday, 0800 – 1600) no less than five days prior to the events.

C.3.092.2.2 [Orig] ATs shall be conducted during normal working hours (Monday – Friday, 0800 – 1600) unless specifically required to be conducted during other times or coordinated and agreed to by the USCG.

C.3.092.2.3 [Orig] Notice of AT cancellation shall be provided to the USCG at least 48 hours in advance of the scheduled test date. Cancelled events shall not be restarted without notifying the USCG at least 72 hours in advance unless otherwise approved by the USCG.

C.3.092.3 [Orig] Test Documentation

C.3.092.3.1 [Orig] The Contractor shall prepare the following documentation in support of the AT program:

C.3.092.3.1.1 [Orig] AT Plan – The Contractor shall develop an AT Plan which describes roles and responsibilities, when, where, and how ATs will be conducted to verify that the boat performance criteria have been met, and that all equipment operates properly. The plan shall include the traceability between the specification and test procedures. The traceability will have a unique identifying number. The plan shall also describe how each AT procedure will be documented, how results will be recorded, and the format for the final report for each boat. The plan shall include a sample AT report. The plan may outline how test events will be tailored to consolidate testing requirements provided that all test requirements are met. **[CDRL]**

C.3.092.3.1.2 [Orig] AT Procedures – The Contractor shall prepare AT Procedures for each test to be conducted. The procedures shall fully describe the test and include a description of how the boat will be operated during the test/inspection, and any test limitations imposed. The AT procedure shall include all pass/fail criteria, a list of testing equipment, performance data to be recorded, step-by-step operations to obtain the required data, and the unique identifying number. Procedures shall be in sufficient detail to allow the USCG to duplicate the test and results. ATs shall not be conducted until the AT Procedures are accepted by the USCG in accordance with Appendix A. Deviations and changes from an accepted procedure or an accepted AT sequence shall be prohibited. Except as required by the AT Procedure, adjustments on equipment being tested shall not be made during the testing. Any adjustment shall require a retest. For results to be valid, each test event shall be conducted in its entirety, per the USCG accepted AT Procedures. **[CDRL]**

C.3.092.3.1.3 [Orig] AT Schedule – The AT Schedule shall indicate the scheduled dates for all AT Procedures. **[CDRL]**

C.3.092.3.1.4 [Orig] AT Reports – The AT Reports shall document the conduct of the test. The report shall include records of any measurements or data taken during the testing, identify any deficiencies, describe the corrective actions to be taken, indicate the test has passed the criteria, and has been signed by the Contractor administering the AT and the USCG, if present. **[CDRL]**

C.3.092.3.1.5 [Orig] Failure and Discrepancy Reports – In the event of any failure to pass the acceptance criteria established in any procedures, the Contractor shall prepare and submit a failure and discrepancy report for each failure. The report shall provide an analysis of the failure, reason for unsuccessful performance, corrective action taken, and any supplemental information related to the failure. The Contractor shall remain responsible for schedule impact due to the nonconformity and shall be subject to appropriate action by the Contracting Officer including a stop work order. **[CDRL]**

C.3.092.3.2 [Orig] For each underway test the following information shall be recorded in addition to the data collected for the test:

C.3.092.3.2.1 [Orig] Fuel – Amount of fuel onboard at the start and end of the test.

C.3.092.3.2.2 [Orig] Personnel – Number of persons onboard.

C.3.092.3.2.3 [Orig] Equipment Weights – Weights and centers of gravity of any equipment carried onboard that is not part of the loading condition outfit.

C.3.092.3.2.4 [Orig] Weather and Sea Conditions – Taken at the beginning and end of each run. The conditions taken shall include wind speed and direction, air temperature, water temperature, and wave height and direction.

C.3.092.4 [Orig] Underway Operations

C.3.092.4.1 [Orig] Anytime a boat is underway for any reason, it shall be operated by a captain who possesses a current valid Coast Guard license for the operation of vessels up to 100 tons, in the waters in which the boat will be operated. In addition, the following shall be required for every underway operation:

C.3.092.4.1.1 [Orig] At least two additional crew members shall be onboard who are able to assist in operating the boat.

C.3.092.4.1.2 [Orig] The Captain shall have onboard a safety kit that, at a minimum, includes a portable VHF-FM radio, a portable GPS navigation device, a 10- person first aid kit, and an appropriate set of flares for the waters being navigated.

C.3.092.4.2 [Orig] Only the National Ensign may be displayed or flown while underway.

C.3.092.4.3 [Orig] Immediately prior to getting underway, the Captain shall notify the local USCG sector via VHF-FM radio with the following information:

C.3.092.4.3.1 [Orig] Hull identification number

C.3.092.4.3.2 [Orig] The number of Contractor personnel onboard

C.3.092.4.3.3 [Orig] The number of USCG personnel onboard

C.3.092.4.3.4 [Orig] The planned area of operations

C.3.092.4.3.5 [Orig] The estimated time of return

C.3.092.4.4 [Orig] If the boat transits from one sector's area of responsibility (AOR) to another, the Captain shall notify the new Sector.

C.3.092.4.5 [Orig] Upon mooring, the Captain shall notify the Sector that the boat is no longer underway.

C.3.092.5 [Orig] Minimum Testing On All Boats

C.3.092.5.1 [Orig] Watertight integrity testing

C.3.092.5.1.1 [Orig] Conduct watertightness test for degree 2 and 3 components in accordance with ISO 12216-D.2.

C.3.092.5.1.2 [Orig] Conduct watertightness test for degree 1 components via one of the following methods:

C.3.092.5.1.2.1 [Orig] After installation by continuous immersion of 1 inch of water for 3 minutes with no leakage. A temporary cofferdam may be used to bound the test area and avoid immersing components not undergoing the test.

C.3.092.5.1.2.2 [Orig] A water test conducted in accordance with ISO 12216-D.2, with the exception that the criteria for a successful test shall consist of no evidence of water on the opposite side of the structure.

C.3.092.5.1.2.3 [Orig] A chalk test for doors or hatches. Prior to testing ensure that the door or hatch has been properly adjusted. Chalk the bearing surface or knife edge and close the door or hatch by normal procedures. When the door or hatch is opened, the chalk from the knife edge will have been transferred to the gasket. A successful test is noted by a uniform and continuous chalk mark on the door's or hatch's gasket (100 percent gasket contact). Irregularities or breaks in the chalk mark are cause for failure.

C.3.092.5.2 [Orig] Propulsion system

C.3.092.5.3 [Orig] Propulsion controls

C.3.092.5.4 [Orig] Propulsion emergency shutdown system

C.3.092.5.5 [Orig] Steering controls

C.3.092.5.6 [Orig] Steering system

C.3.092.5.7 [Orig] Bow Thruster (if applicable)

C.3.092.5.8 [Orig] Gauges and alarms

C.3.092.5.9 [Orig] Bilge Drainage system

C.3.092.5.10 [Orig] Electrical system

C.3.092.5.11 [Orig] Starting system

C.3.092.5.12 [Orig] Electronic systems

C.3.092.5.13 [Orig] Communication system

C.3.092.5.14 [Orig] Navigation lights

C.3.092.5.15 [Orig] Lighting systems

C.3.092.5.16 [Orig] HVAC system

C.3.092.5.16.1 [Orig] System testing and balancing shall be in accordance with ASHRAE Standard 151.

C.3.092.5.17 [Orig] Window wipers and washing

C.3.092.5.18 [Orig] Fire suppression system

C.3.092.5.19 [Orig] Potable water system

C.3.092.5.20 [Orig] Sewage system

C.3.092.5.21 [Orig] Greywater system

C.3.092.5.22 [Orig] ATON Crane system

C.3.092.5.22.1 [Orig] The crane shall be tested by applying an external force for 10 minutes, to demonstrate structural adequacy of the equipment and foundation. The test force shall be applied in the weakest direction identified within the normal arc of loading for each fitting and identified in the test procedure. No part of the equipment, fittings, and structure shall take a permanent set, nor shall degradation to any operating or control function occur because of the test. Testing load shall be 800 lbf.

C.3.092.5.23 [Orig] Anchor handling

C.3.092.5.24 [Orig] Cross Deck Winches

C.3.092.5.25 [Orig] Washdown system

C.3.092.5.26 [Orig] Demonstrate Start-Up Procedures – The Contractor shall demonstrate and perform start-up procedures while the boat is at the dock before getting underway.

C.3.092.5.27 [Orig] Full Power and Endurance Test – The boat shall undergo a two-hour full power and endurance test. Full power shall be demonstrated for one hour. The endurance test shall be performed at varying speeds at or above the cruising speed for at least one hour. During this test the Contractor shall demonstrate that all parts of the propulsion system, as well as all other equipment and systems are in satisfactory operating condition. The Contractor shall conduct inspections to verify that there are no leaks in any piping system. The Contractor shall record readings on all gauges and meters in 10-minute intervals. This test shall not precede any required engine break-in requirements.

C.3.092.5.28 [Orig] Locked Shaft Test – The shaft holding mechanism for one shaft shall be engaged while stationary. Power shall then be applied to the free shaft gradually increasing to full throttle. Return to zero, disengage the shaft holding mechanism, and then repeat for the other shaft.

C.3.092.5.29 [Orig] Speed Test – The Contractor shall undergo speed trials to demonstrate the boat will achieve the full speed. This speed test may be performed in calm conditions or above, provided that the full speed can be attained. OEM-required break in periods shall be conducted prior to speed tests. The speed tests shall be comprised of two runs, in opposite directions over a one nautical mile course, at a minimum, with the speeds for the two runs averaged. The Contractor shall record readings on all gauges and meters for each run.

C.3.092.5.30 [Orig] Steering Test – The satisfactory operation of the steering system and stability of the boat during turns shall be demonstrated. Test runs shall be made at cruising and full speed. Helm angles used shall be 35 degrees. Any indications of loss of control, failure to maintain a steady turn, or unpredictable motions shall be recorded.

C.3.092.5.31 [Orig] Range Verification – The Contractor shall provide verification that the boat will achieve the range requirement.

C.3.092.5.32 [Orig] Maneuvering Testing – The Contractor shall conduct testing to demonstrate the boat meets the USCG operating maneuverer requirements such as, quick throttle reversals, holding station, and backing. The maneuvering testing shall be conducted across all speeds.

C.3.092.5.33 [Orig] Post-Underway Maintenance – The Contractor shall perform post-underway shut-down procedures and checks in accordance with the BIB.

C.3.092.5.34 [Orig] Scale Weighing – A scale weighing shall be conducted on all boats at the defined Lightship Condition. The Contractor shall include in the scale weighing report the procedure used to determine Lightship Weight.

C.3.092.5.34.1 [Orig] The scale weighing shall use a minimum of two lift points or load cells. The as-built Longitudinal Center of Gravity (LCG) shall be calculated from the scale weighing.

C.3.092.5.34.2 [Orig] The weight of any follow-on hull cannot vary from the weight of the current configuration baseline by more than two percent, unless previously authorized by the Coast Guard. In the event the weight has increased outside the allowable limits, a Failure and Discrepancy Report shall be prepared. The report shall address the following:

C.3.092.5.34.2.1 [Orig] Reason for unsuccessful performance;

C.3.092.5.34.2.2 [Orig] Any supplemental information related to the failure;

C.3.092.5.34.2.3 [Orig] Corrective action; and

C.3.092.5.34.2.4 [Orig] Impact to schedule.

C.3.092.5.34.2.5 [Orig] Any corrective adjustments to the documented weight shall be made in accordance with CM procedures.

C.3.096 [Orig] Weight Control and Weight Reports

C.3.096.1 [Orig] General

C.3.096.2 [Orig] The Contractor shall follow weight control recommendations as outlined in Society of Allied Weight Engineers (SAWE RP-14), 2001; Recommended Practices 14, Weight Estimating and Margin Manual for Marine Vehicles.

C.3.096.3 [Orig] In determining the loading conditions and seaworthiness,

C.3.096.3.1 [Orig] Crew's and passenger's weight shall be 220 pounds per person.

C.3.096.3.2 [Orig] All crews and passengers aboard shall be seated with a VCG 2.5 inches above the surface they are seated on.

C.3.096.4 [Orig] Weight Control Plan

C.3.096.4.1 [Orig] The Contractor shall develop and implement a weight control plan for the boat that includes a description of the Contractor's procedures for determining weights and controlling weight growth during Design and Development.

C.3.096.4.2 [Orig] The objective of the weight control plan is to ensure that the final weight and centers of gravity of the boat meet the system and seaworthiness requirements.

C.3.096.4.3 [Orig] The weight control plan shall address the following topics at minimum, and specific guidance provided shall be incorporated:

C.3.096.4.3.1 [Orig] Monitoring and updating weight data with progress of overall design and development.

C.3.096.4.3.2 [Orig] Address use of the limiting plots/tabular values developed in stability analysis to evaluate trends of the previous and most recent weight estimates vs. limits for compliance with the stability and floodable length requirements.

C.3.096.4.3.3 [Orig] The action that will be taken upon detection of weight or centers trends tending towards not meeting the system requirements and seaworthiness.

C.3.097 [Orig] One-Time Tests (OTTS)

C.3.097.1 [Orig] General

C.3.097.1.1 [Orig] The Post Construction Inspection and one-time tests shall be conducted once on the first boat of the first delivery order and only repeated on an as-needed basis.

C.3.097.1.2 [Orig] Unless otherwise specified, the one-time tests (OTTS) shall be included in the AT plan. These tests' test procedures shall be developed and results shall be reported as per AT Reports.

C.3.097.1.3 [Orig] One-time tests shall be performed at the Contractor's facility or at a location on the U.S. East Coast as agreed upon by both parties.

C.3.097.1.4 [Orig] Notice of cancellation of any inspection or test shall be provided to the USCG at least 48 hours in advance of the scheduled date. Cancelled events shall not be restarted without notifying the USCG at least 72 hours in advance unless otherwise approved by the USCG.

C.3.097.2 [Orig] Inclining Experiment

C.3.097.2.1 [Orig] The boat shall be inclined in accordance with ASTM F1321-2 to determine the weight and center of gravity.

C.3.097.2.2 [Orig] At least three mechanisms to measure heel angle shall be used.

C.3.097.2.3 [Orig] At least one pendulum or water level shall be used in addition to any other means of measuring heel angle, such as electronic inclinometers.

C.3.097.2.4 [Orig] The Contractor shall prepare an Incline Test Plan that describes how the incline test will be completed, how the boat will be supported, what weights will be used and their placement, and how measurements are taken and recorded. All data sheets to be used in the inclining experiment shall be included in the Incline Test Plan. **[CDRL]**

C.3.097.2.5 [Orig] The boat shall be inclined with the fuel tank(s) and any other spaces that might contain liquids, including the bilges, emptied.

C.3.097.2.6 [Orig] All inclining weights, scales, equipment for observations, cribbing, and other material required for the test shall be furnished by the Contractor. The Contractor shall also provide all labor necessary for preparing the boat for the incline, installing apparatus, taking measurements and observations, handling lines, and shifting incline weights during the test.

C.3.097.2.7 [Orig] The Contractor shall prepare an Incline Test Report (in the Contractor's format) presenting the description and results of the incline test. **[CDRL]**

C.3.097.2.8 [Orig] The report shall include an error analysis determining the 95 percent confidence level accuracy of the data for displacement, moment-tangent slope, and of the resultant height of center of gravity above baseline or keel (KG).

C.3.097.3 [Orig] Speed, Power, and Fuel Consumption Characteristics Test **[CDRL]**

C.3.097.3.1 [Orig] The speed, power, and fuel consumption characteristics shall be demonstrated and recorded during this test. Speed runs shall be made in water at least 30 feet deep. At least 2 runs, 1 in each direction, shall be made for each speed over a minimum of 2 nautical miles and the results averaged. At least 6 speeds, evenly distributed from idle to full speed, shall be used to prepare graphs of speed vs. engine RPM, engine percent load vs. engine RPM, fuel consumption vs. speed, fuel consumption vs. RPM, and running trim vs. speed. These trials shall be conducted in accordance with the test plan. This test may be performed in calm water condition, or above, as agreed on by the Contractor and the USCG. The test shall be conducted in three weight conditions: full load ATON, minimum operating, and a 50% fuel condition.

C.3.097.3.2 [Orig] The Contractor shall measure and record all installed gauges and meter readings on each engine at speeds evenly distributed from idle to full. The measurements shall be taken once each data point has been reached and stable for five minutes. The parameters shall be within acceptable ranges as specified by the OEM.

C.3.097.4 [Orig] Bollard Pull Test **[CDRL]**

C.3.097.4.1 [Orig] The Contractor shall conduct a bollard pull test on the boat to determine the maximum available static bollard pull at full load icebreaking. The bollard pull test shall be conducted in accordance with the International Standard for Bollard Pull Trials prepared by: Bollard Pull Trials Industry Project Partners, MARIN 2019, with the following exceptions:

C.3.097.4.2 [Orig] If engine speed cannot be recorded digitally it may be recorded manually from the boat's gauges or displays as frequently as practicable.

C.3.097.4.3 [Orig] Engine torque and power measurements are not required.

C.3.097.5 [Orig] Noise Test **[CDRL]**

C.3.097.5.1 [Orig] The Noise Test shall be conducted underway, at the full load ATON, and test the following operating conditions: idle and up to cruise speed.

C.3.097.5.2 [Orig] The Contractor shall prepare a Noise Test Report detailing the test performed and its results.

C.3.097.6 [Orig] Lighting Survey **[CDRL]**

C.3.097.6.1 [Orig] The Contractor shall develop a test procedure and conduct a lighting survey to demonstrate all lighting system illuminance levels.

C.3.097.7 [Orig] Maneuvering Tests **[CDRL]**

C.3.097.7.1 [Orig] The Contractor shall develop a test procedure and demonstrate the HSC-L maneuvering performance in accordance with the procedures stated in ABS Guide for Vessel Maneuverability.

C.3.097.7.2 [Orig] The Contractor shall prepare a Maneuverability Test Report detailing the test performance and its results.

C.3.097.8 [Orig] Tow Bitts and Mooring Fittings Pull Test **[CDRL]**

C.3.097.8.1 [Orig] The mooring, towing, and anchoring fittings shall be tested by applying an external force for 10 minutes to demonstrate structural adequacy of the equipment and foundation. The test force shall be applied in the weakest direction identified within the normal arc of loading for each fitting and identified in the test procedure. No part of the equipment, fittings, and structure shall take a permanent set, nor shall degradation to any operating or control function occur because of the test. Testing load shall be at the rated strength.

C.3.098 [Orig] Mockups

C.3.098.1 [Orig] General

C.3.098.1.1 [Orig] The Contractor shall construct physical mockups of the pilothouse, and aft deck on the boat for physically evaluating the arrangements.

C.3.098.1.2 [Orig] A 3D model of the relevant space shall be provided in advance of Mockup Reviews. The mockup evaluation events will be in alignment with the design review process.

C.3.098.1.3 [Orig] All mockups shall be at the Contractor's facility.

C.3.098.1.4 [Orig] The Contractor shall request Coast Guard subject matter experts (e.g., active-duty personnel) to support testing and evaluation using the mock-up.

C.3.098.1.5 [Orig] The USCG will use the 3D model and mockup to evaluate operations by the crew, access to controls, range of visibility from seated and standing positions, accessibility to equipment, and all aspects of operations from the pilothouse, and aft deck. The mockups shall be maintained and made available through the completion of OT&E to support use by the Contractor and the USCG in the development of engineering changes identified through the OT&E.

C.3.098.1.5.1 [Orig] Usability Testing shall be defined according to ISO 9241-11:2018: Usability is the "extent to which a system, product, or service can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use."

C.3.098.1.6 [Orig] Prior to disassembly, the contractor shall consult with the HSC-L PRO for items the USCG desires to retain, and following disassembly, deliver any requested components to the HSC-L PRO at the Production Facility.

C.3.098.2 [Orig] Mockup Requirements

C.3.098.2.1 [Orig] The Contractor shall design and construct the mockups using MIL-M-23530A Sections 3.1, 3.2, 3.4, and 3.8 as guidance. Mockups for the boat shall be in accordance with MIL-M-23530A, Sections 1.2, Type II (full scale), and

C.3.098.2.1.1 [Orig] Class 1 (exact reproduction of shape and external dimensions) for

C.3.098.2.1.1.1 [Orig] FWD helm station control surfaces; and

C.3.098.2.1.1.2 [Orig] AFT helm station control surfaces; and

C.3.098.2.1.1.3 [Orig] Crane operator station; and

C.3.098.2.1.1.4 [Orig] seats

C.3.098.2.1.2 [Orig] Class 2 (reproduction of general shape and major external dimensions in sufficient detail to permit easy identification and to ensure arrangements compatibility with all adjacent equipment, components, and systems) for

C.3.098.2.1.2.1 [Orig] displays; and

C.3.098.2.1.2.2 [Orig] Aft deck (this includes the towing tackle arrangement, crane, winches).

C.3.098.2.2 [Orig] All mockup components shall be labeled.

C.3.098.2.3 [Orig] Non-functioning mockups of individual components may be used in all spaces, unless otherwise noted in this section, provided that all relevant functions and access points are conveyed.

C.3.098.2.4 [Orig] The mockup shall be reconfigurable for demonstration and evaluation purposes.

C.3.098.2.5 [Orig] The mockups shall correctly reflect space and volume, egress passageways and aids (handholds, rails, steps, etc.), arrangements, location, and layout of controls of the represented space or area. The mockup does not need to reflect noise, vibration, climate, or environmental conditions that the boat will experience. Mockups of spaces in the open air environment shall include the first overhead obstructions within eight vertical feet of the mockup deck.

C.3.098.2.6 [Orig] The Pilothouse and Aft Deck mockups shall be co-located for mockup evaluations and oriented and arranged spatially to each other reflective of the actual design.

C.3.098.3 [Orig] Mockup Materials

C.3.098.3.1 [Orig] The mockup shall be constructed of suitable durable materials (e.g., plywood, light-gauge aluminum) that will facilitate easy reconfiguration of boat systems during the design review processes.

C.3.098.3.2 [Orig] Actual seats of the model/type planned for HSC-L shall be used in the mockup to demonstrate clearances, reach, and mounting.

C.3.099 [Orig] Photographs and Videos

C.3.099.1 [Orig] The Contractor shall allow the Government to take photographs and video in the yard and onboard the boat for Government use. The Contractor shall establish a process for the Government to take photographs and videos.

C.3.099.2 [Orig] Photographs and videos of the HSC-L taken by the Contractor shall be made available to the Government and shall not be restricted by copyright, patent, or any other restraint in the use or disposition of any files or prints.

C.3.099.3 [Orig] The Contractor shall take identification photographs when the boat is on trials. Photographs shall show the boat with staging and other extraneous equipment removed to ensure a true representation of the boat as it will appear when ready for sea.

APPENDIX A: [ORIG] CDRL MILESTONE CHART

To Be Developed

For guidance[1]:

1 The CDRL Milestone Chart is provided for reference and guidance only. Refer to the CDRLs for submission requirements.

APPENDIX B: [ORIG] LIST OF REFERENCES

To be Developed

APPENDIX C: [ORIG] LIST OF GOVERNMENT FURNISHED INFORMATION

To be Developed